

Individual dopaminergic neurons induce unique, yet overlapping combinations of behavioural modulations including safety learning, memory retrieval and acute locomotion

Naoko Toshima, Arman Behrad, Franziska Behnke, Gauri Kaushik, Aliće Weiglein, Martin Strauch, Juliane Thoener, Oliver Kobler, Maia Lisandra M Wang, Markus Dörr, Michael Schleyer 

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany • Hokkaido University, Institute for the Advancement of Higher Education, Sapporo, Japan • Hokkaido University, Graduate School of Life Science, Sapporo, Japan • RWTH Aachen University, Imaging and Computer Vision, Aachen, Germany • Leibniz Institute for Neurobiology (LIN), Combinatorial NeuroImaging Core Facility, Magdeburg, Germany • Hokkaido University, School of Science, Sapporo, Japan

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study presents **valuable** findings on the role of specific dopamine neurons for aversive learning and modulation of innate behavior in *Drosophila* larvae. The authors present **solid** evidence backed up by detailed behavioral quantification and rigorous testing. Their data confirms previous findings and will be of interest to the learning and memory community.

<https://doi.org/10.7554/eLife.106148.1.sa3>

Abstract

Two evolutionary highly conserved functions of dopamine are to carry “teaching” signals during associative learning and to control movement. In mammals and humans these functions are generally thought to be produced by different populations of neurons. Here, we investigated in the larva of *Drosophila melanogaster* whether both these functions can be induced by the same individual dopaminergic neurons in the central brain. Focusing on the dopaminergic neurons of the DL1-cluster, we asked whether the optogenetic activation of individual neurons established associative punishment and/or safety memories, controlled the retrieval of the established memories, and acutely modulated locomotion. We found that each neuron had a unique, yet overlapping set of behavioural effects. Several individual neurons both established a memory and modulated acute locomotion by increasing the animals’ bending and decreasing its velocity. Our results demonstrate that individual dopaminergic neurons can fulfil a surprisingly broad range of functions in different

behavioural contexts. Given the highly conserved roles of the dopaminergic system across the animal kingdom, this study raises the question whether a similarly diverse functionality can be found also in other animals, including humans.

Introduction

Dopamine has many functions in the brain. Among them, two turned out to be highly conserved throughout the animal kingdom: to carry “teaching” signals during associative learning and to initiate and control movement^{1–5}. In mammals and humans, it is widely accepted that these two functions are brought about by different populations of dopaminergic neurons (DANs): effects regarding learning are thought to be elicited by the mesolimbic pathway, effects regarding movement by the nigrostriatal pathway⁴. Here, we study in the larva of the fruit fly *Drosophila melanogaster* whether in contrast the same individual central brain-DANs can induce both functions regarding memory and movement.

The fruit fly larva is an established model organism to study the neuronal and genetic mechanisms underlying associative learning^{6–7}. Due to the numerical simplicity of its brain and the available toolbox for transgene expression, the larva allows investigating these mechanisms at tractable levels of complexity. In *D. melanogaster* distinct sets of DANs convey appetitive and aversive unconditioned stimuli (US), respectively (larvae^{6–7}; adults^{2,8}) (a similar scenario may be emerging in vertebrates, too^{1,9,10–13}). Each of the DANs synapses onto a different compartment of the parallel fibres of Kenyon cells (KCs) that form a higher-order brain structure called mushroom body (larvae⁷; adults¹⁴). Odours are combinatorically coded across the array of these KCs which in turn give output to mushroom body output neurons (MBONs). The MBONs collect information across the KCs and send it towards efferent circuitry, promoting either approach or avoidance (larvae⁷; adults¹⁴). Within the KCs the coincidence of odour presentation and activity in DANs is detected and enables a memory trace to be formed. The respective memory engram is thought to come by synaptic depression between the KCs and the MBONs of the same mushroom body compartment, such that the net effect of the activity across the MBONs is tilted towards approach or avoidance, respectively (larvae^{15–16}; adults^{17–18}). Electron microscope reconstructions further revealed a direct within-compartment connection from the DANs onto the MBONs (larvae¹⁹; adults²⁰). Thus, each DAN has two main targets: the KCs and the MBONs of its own compartment. The general organization of this DAN-KC-MBON matrix is likely shared with other insects^{2,21–23}.

Three pairs of identified DANs of the dorso-lateral cluster 1 (DL1) have been shown to support aversive learning upon optogenetic activation during Pavlovian conditioning^{15,24–25}. In this study, we ask whether those same DANs have additional functions. Specifically, we ask for their role for safety learning – that is, learning that a cue predicts the absence of a threat; memory retrieval – that is, whether a memory will be retrieved and affect behaviour; and modulations of acute, innate locomotion. Little is known about how the dopaminergic system controls any of these functions – if at all. Given the highly conserved functionality of the dopaminergic system across the animal kingdom, this study will reveal more about how the manifold functions of the dopaminergic system are organized – in any brain.

Results

We focus our study on the DANs that support aversive learning when optogenetically activated, called DAN-d1, DAN-f1 and DAN-g1. We investigated the transgenic expression pattern of the driver strains covering each DAN, and the effects of activating each DAN on learning, memory retrieval and locomotion.

The used driver strains drive transgene expression in identified DANs

Previous studies had described the expression patterns of the used split-driver strains in the central nervous system in detail^{15,25,27}. Here, we extended the view on the expression within the whole body. Towards this end, we applied a recent protocol for whole-body visualizations²⁸. We confirmed the highly specific transgene expression in the brain of the four split-Gal4 driver strains without any detectable expression in the rest of the body (**Fig. 1A-D**, Supplementary material S1). *TH-Gal4* is known to cover almost all DANs in the brain, with the notable exception of the neurons of the primary protocerebral anterior medial (pPAM) cluster²⁶. We confirm this pattern of transgene expression and in addition found strong expression outside of the brain, in particular in the proventriculus and in hypodermal melanin-producing cells (**Fig. 1E**, Supplementary Movie 1). This pattern fits to the reported function of dopamine in the hardening and darkening of the insect cuticle^{29–30} and a recent report of *TH-Gal4* expression in the adult proventriculus³¹. This result means that any genetic manipulation using *TH-Gal4* may affect cells outside of the central nervous system. Finally, we confirm that *R58E02-Gal4* covers three of the four pPAM DANs that are missing in the *TH-Gal4* expression pattern²⁷, with no expression outside of the central nervous system (**Fig. 1F**).

Some individual DANs establish and retrieve associative memories

In larvae, for many kinds of aversive US learned behaviour is only observed if the US is present at the moment of memory retrieval^{32–36}. For example, we trained groups of larvae with an odour as the conditioned stimulus (CS) and quinine as US and observed an aversive associative memory when we tested the animals in presence of quinine, but not in its absence (**Fig. 2A**). Learned aversive behaviour therefore can be viewed as an escape from the US, guided by the respective memory – an escape that is pointless in absence of anything to escape from^{37–40}.

We then asked for the roles of the DL1-DANs for both establishing and retrieving associative memories by employing the same learning experiment, but instead of quinine we used optogenetic DAN activation by blue light as a virtual US. We also tested the innate odour preference of all genotypes, both in darkness and in blue light (**Fig. 2 – Figure supplement 1**) and observed no evidence that the innate odour preference was modified by activating the respective DANs.

After training with DAN-d1 as virtual US, zero memory scores were observed when the animals were tested in darkness. When tested in blue light, however, the animals of the experimental genotype, but not the controls, showed aversive associative memory (**Fig. 2B**, **Fig. 2 – Figure supplement 2A-B**). Interestingly, a recent study using a much stronger light intensity during training observed no significant memory with DAN-d1 as virtual US²⁵. We therefore repeated the experiment with two different light intensities and observed an aversive memory only when using the low intensity that we use throughout the study, but not with a 10-fold higher intensity (**Fig. 2 – Figure supplement 2C**). Moreover, when raising the larvae in food supplemented with all-trans-retinal which had been shown to increase the effectiveness of our effector ChR2-XXL⁴¹, no aversive memory could be detected (**Fig. 2 – Figure supplement 2D**). Thus, it seems that a too strong activation of DAN-d1 prevents the establishment and/or retrieval of an aversive memory.

For DAN-f1, we likewise found an aversive memory that was retrieved only when DAN-f1 was activated also during the test (**Fig. 2C**, **Fig. 2 – Figure supplement 2E**). Our results indicate that both DAN-d1 and DAN-f1 are sufficient to establish an aversive memory which requires the respective DAN to be activated for retrieval.

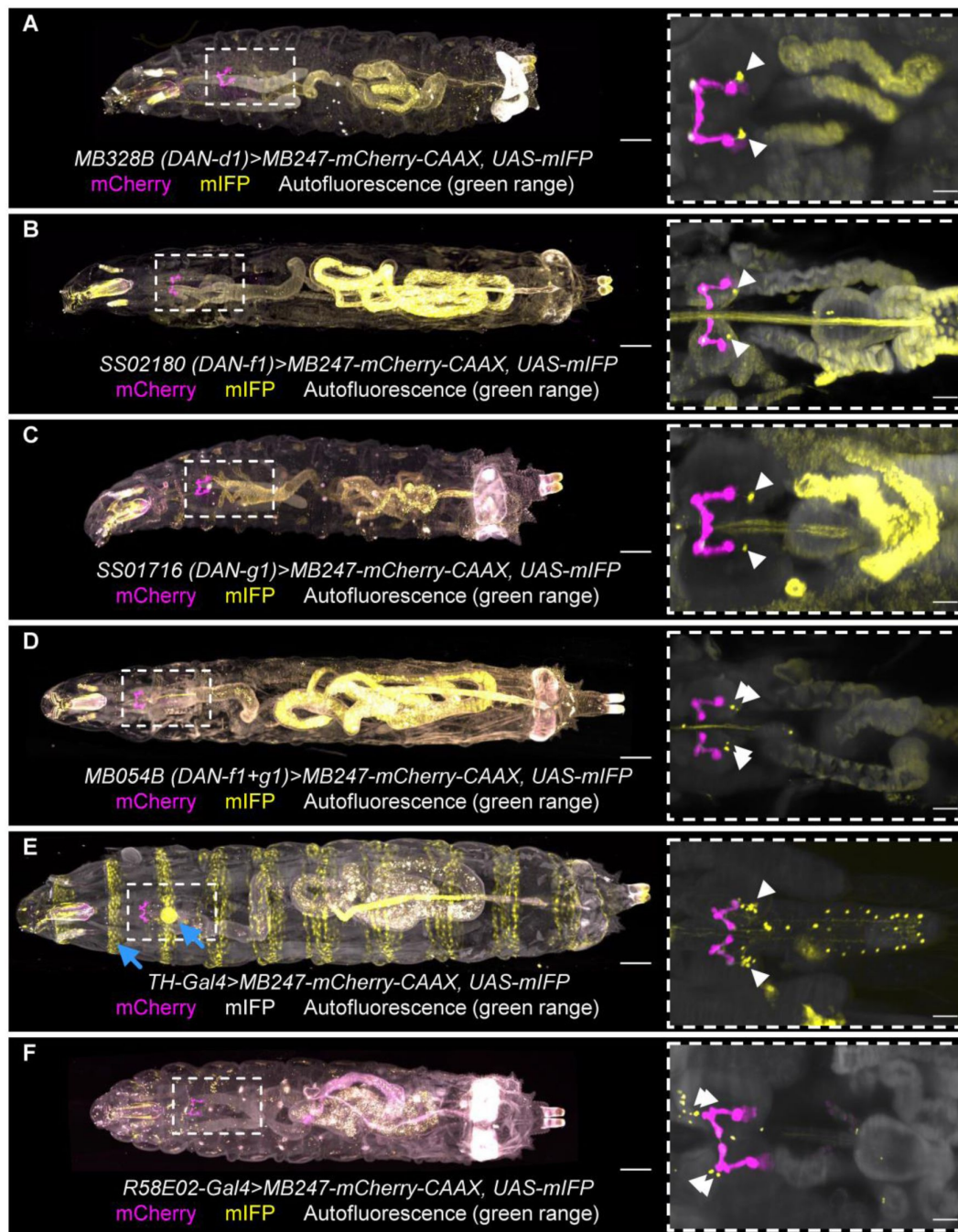


Figure 1

Whole-body expression patterns of DAN-driver strains.

All driver strains were crossed to a MB247-mCherry-CAAX, UAS-mIFP double effector that will show a mCherry signal (magenta) in the mushroom body and a mIFP signal (yellow) controlled by the driver strain. In addition, autofluorescence in the green range is shown in white. The insert shows a volume-restricted magnification of the CNS region. For images of each separate channel, as well as for genetic controls, see supplementary material S1. (A) MB328B, (B) SS02180, (C) SS01716 and (D) MB054B each drive expression in one to three cells per hemisphere (arrowheads) innervating the mushroom body as described previously^{15,25,25}. No additional cells were found in the body. (E) TH-Gal4 drives additionally to the expression in the CNS (arrowheads indicate the DL1 cluster) also outside of the brain, most prominently in the melanin-producing cells at the segment barriers and the proventriculus (blue arrows). (F) R58E02-Gal4 drives expression in three cells per hemisphere (arrowheads) innervating the medial lobe of the mushroom body, as previously described^{27,47}. Scale bars represent 250 μm for the whole body and 50 μm for the inserts.

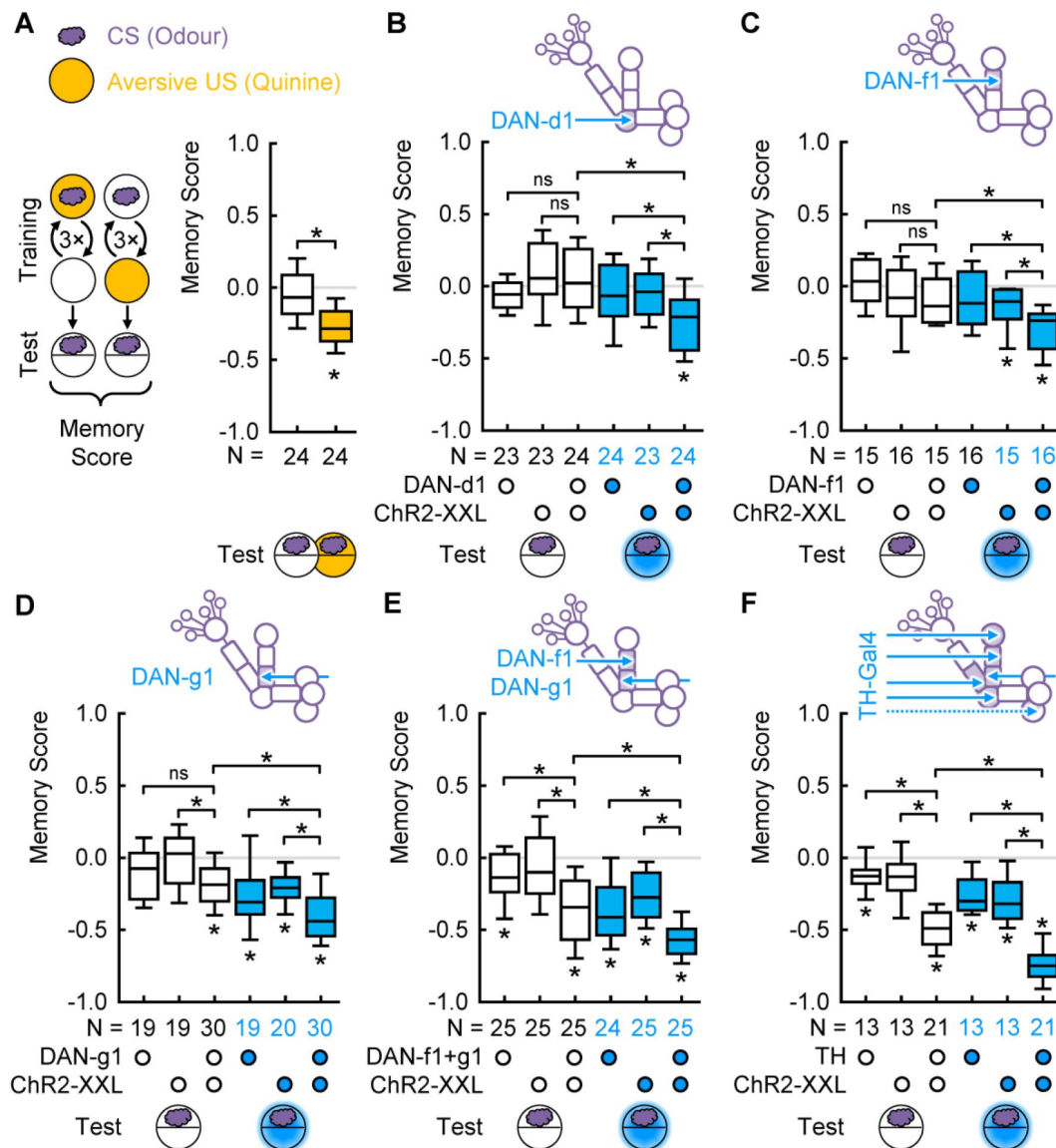


Figure 2

Individual DANs are sufficient to establish and retrieve aversive memory.

(A) After training wild-type larvae with quinine as US, associative memory is observed only in presence of the US but not in its absence. (B) When animals were trained with DAN-d1 activation as US and tested in darkness, memory was retrieved in neither genotype. When tested in presence of blue light, negative memory scores only in the experimental genotype indicated aversive associative memory. For independent repetitions, see Figure supplement 2A-D. (C) As in (B), but with DAN-f1 activation. For an independent repetition, see Figure supplement 2E. (D) In the case of DAN-g1, aversive memory scores were observed in the experimental genotype even in darkness. When tested in blue light, aversive memory scores were observed in all genotypes, but significantly stronger in the experimental genotype. (E) As in (D), but with a combination of DAN-f1 and DAN-g1. For independent repetitions, see Figure supplement 2F-G. (F) As in (D), but with the broad *TH-Gal4* pattern. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW), * below boxes indicate memory scores significantly different from zero (WSR). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

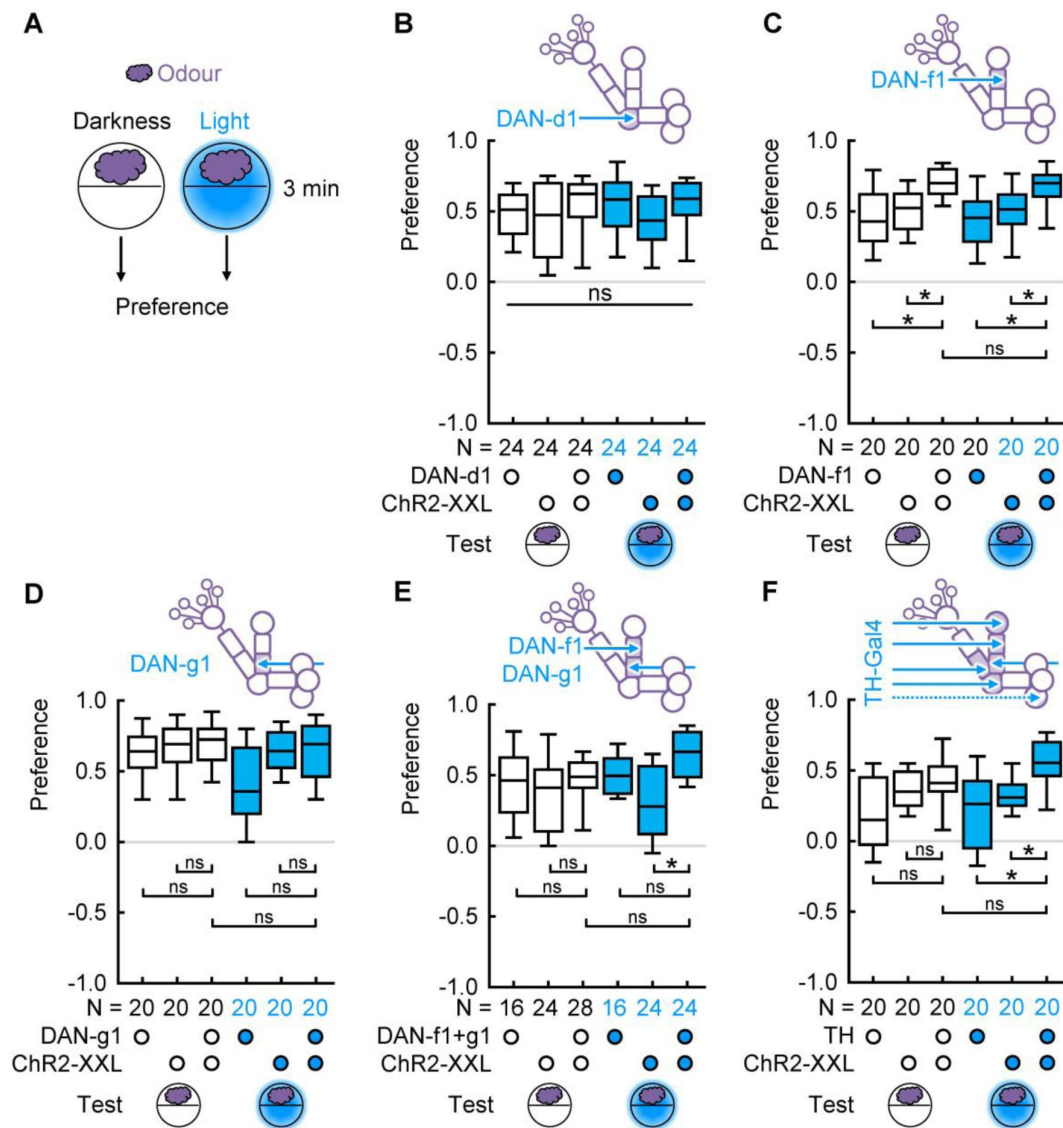


Figure 2 - supplement 1

Innate odour preferences.

(A) Groups of about 20 larvae were tested for three minutes for their odour preference, either in darkness or blue light. (B) Innate odour preferences of the genotypes used in [Fig. 2B](#). Tested in darkness or blue light. No differences were found. (C) Innate odour preferences of the genotypes used in [Fig. 2C](#). The experimental genotypes displayed a higher odour preference than the controls, independent of the testing condition. Note that this cannot explain the difference in memory score between testing conditions found in [Fig. 2C](#). (D) Innate odour preferences of the genotypes used in [Fig. 2D](#). No differences were found. (E) Innate odour preferences of the genotypes used in [Fig. 2E](#). When tested in light, the experimental genotype displayed a higher odour preference than the effector. Note that this cannot explain the difference in memory score between genotypes tested in darkness as seen in [Fig. 2E](#). (F) Innate odour preferences of the genotypes used in [Fig. 2F](#). When tested in light, the experimental genotype displayed a higher odour preference than the controls. Note that this cannot explain the difference in memory score between genotypes tested in darkness as seen in [Fig. 2F](#). ns above a line indicates non-significance across the whole data set (KW), * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

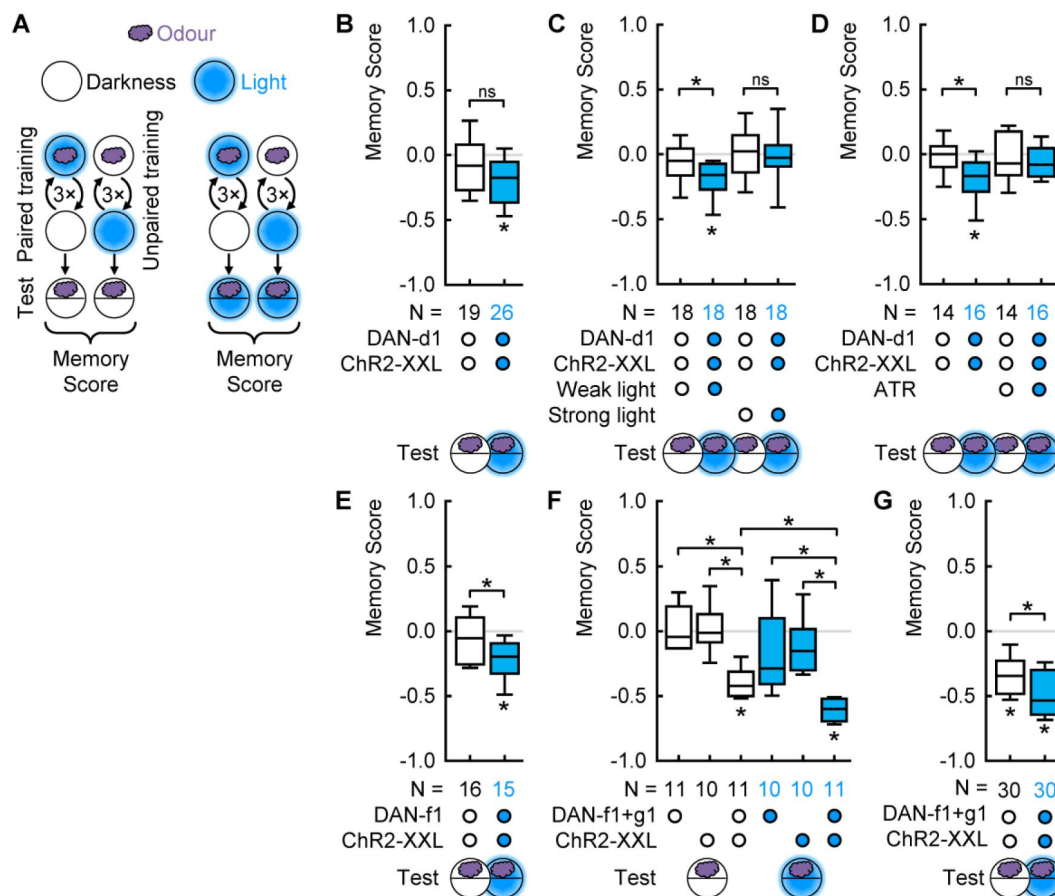


Figure 2 - supplement 2

Replications of the experiments shown in Fig. 2.

(A) Larvae were trained with odour presented either paired or unpaired with blue light activation, and tested either in darkness or in blue light. (B) Larvae of the experimental genotype, expressing ChR2-XXL in DAN-d1, show associative memory only when tested in presence but not absence of the blue light. (C) Larvae of the experimental genotype show associative memory only when trained with the standard weak light and tested in presence of light (of the same strength). When they were trained and tested with a higher light intensity, no memories were observed. (D) Larvae of the experimental genotype show associative memory only when raised in standard food and tested in presence of light. When they were raised in food with all-trans-retinal added, which has been shown to increase the light-sensitivity of ChR2-XXL⁴², no memories were observed. (E) Larvae expressing ChR2-XXL in DAN-f1 show associative memory only when tested in presence but not absence of the blue light. (F) Only the larvae of the experimental genotype, expressing ChR2-XXL in DAN-f1 and DAN-g1, show associative memory independent of the presence or absence of the US. The genetic controls display no significant associative memory. (G) Larvae of the experimental genotype show associative memory only when tested in presence but not absence of the blue light. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW), * below boxes indicate memory scores significantly different from zero (WSR). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

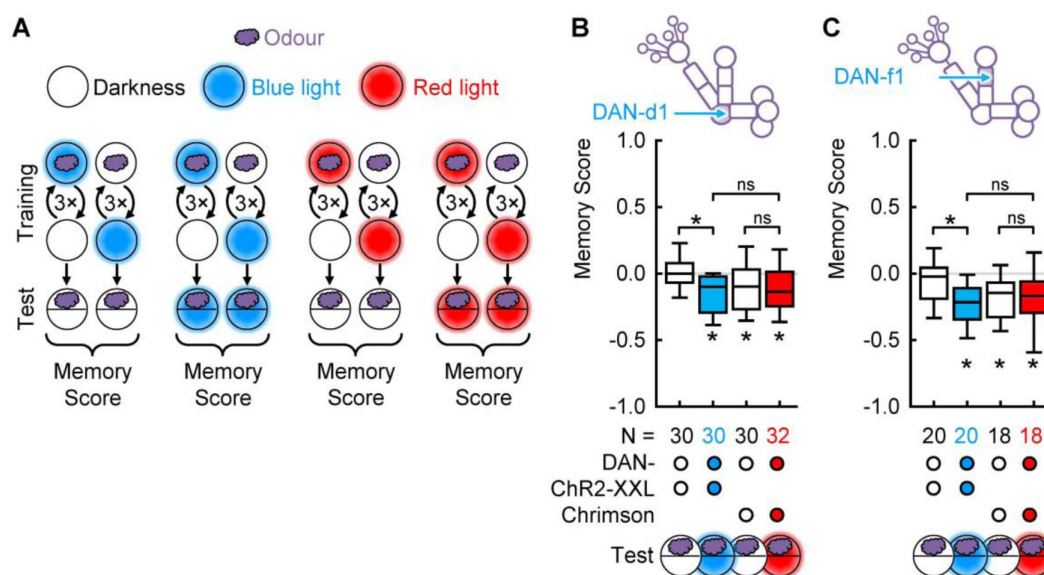


Figure 2 - supplement 3

DAN activation with CsChrimson establishes memories that are retrieved independent of DAN activation during the test.

(A) Larvae were trained with the odour presented either paired or unpaired with blue (*UAS-ChR2-XXL*) or red (*UAS-CsChrimson*) light activation, and tested either in darkness or the respective light. (B) In a replication of the experiment shown in **Fig. 2B**, larvae displayed an associative memory only if DAN-d1 was activated also during the test (left). However, if instead of ChR2-XXL the red-shifted channel CsChrimson was used as effector, associative memory scores were observed independent of the activation of DAN-d1 during the test. (C) As in (B), but for DAN-f1. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW), * below boxes indicate memory scores significantly different from zero (WSR). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

A previous study used the same driver strains crossed to *UAS-CsChrimson* as the optogenetic effector in a similar learning experiment. Critically, learned behaviour was observed also in darkness¹⁵. Using the same effector and the same intensity of red-light stimulation we were able to replicate these results (**Fig. 2 – Figure supplement 3**). Thus, the memories induced by either effector differed in the conditions of how they could be retrieved. These results call for caution against using different optogenetic effectors interchangeably.

Some DANs establish memories but do not control retrieval

Activation of DAN-g1, too, established an aversive memory (**Fig. 2D**). However, in contrast to DAN-d1 and DAN-f1, we found aversive associative memory scores in the experimental genotype independent of the testing condition (**Fig. 2D**).

Next, we employed two driver strains that each covers a combination of the DANs, and activated them using ChR2-XXL. One of them covers DAN-f1 and DAN-g1, the other one, *TH-Gal4*, covers all three of them, plus many additional cells (**Fig. 1E**). In both cases, we observed associative memories in the experimental genotype both in absence and in presence of the US (**Fig. 2E-F**, **Fig. 2 – Figure supplement 2F-G**).

Notably, in the experiments with these driver strains, we also observed significantly aversive memory scores in the genetic controls when tested in light (**Fig. 2D-F**). Given that this result was found both in driver and effector controls, it was most likely caused by the light itself. Light is known as an aversive US in larvae⁴² and has been seen to affect the results of optogenetic experiments before^{24,43}. Why this aversive effect of the light remained a non-significant trend in the previous experiments (**Fig. 2B-C**) is not known.

Taken together, activating DAN-g1, or driver strains that include DAN-g1, established aversive memories that were retrieved independent of whether the respective DANs were activated during the test.

Individual DANs are sufficient for establishing safety, but not punishment memories

Animals can associate stimuli not only with the presence of a threat, but also with its absence⁴⁴. Such safety memory has been found in *D. melanogaster*, but received little attention by the community until recently (larvae^{40,46–48}, adults^{49–50}). That means a US, such as bitter taste, can establish two different memories: punishment memory after paired training and safety memory after unpaired training^{37,46} (**Fig. 3A**). Critically, our results so far demonstrate that each of the individual DANs was sufficient to establish an associative memory – but not, whether this was a punishment memory, a safety memory, or both. Therefore, we next focussed on the animals' odour preference after paired and unpaired training, respectively, compared to the odour preference of the pooled genetic controls as a baseline. A punishment memory would reveal itself by decreased odour preferences after paired training, a safety memory by increased preferences after unpaired training, compared to the baseline. To further bolster this analysis, we included the data of all independent replications of the experiments (**Fig. 2** including Figure supplements 2 and 3).

For the case of DAN-d1, after paired training there was no difference in odour preference between the genetic controls and the experimental genotype, nor between the two testing conditions (**Fig. 3B**, left). After unpaired training, in contrast, the larvae of the experimental genotype exerted increased odour preferences when tested under blue light compared to all other groups (**Fig. 3B**, right). This suggests that the activation of DAN-d1 established no punishment memory, but a safety memory. For DAN-f1, we found analogous results (**Fig. 3C**). In the case of DAN-g1, after paired training we found more negative odour preferences when tested in presence than in absence of light, but no difference between the controls and the experimental genotype (**Fig.**

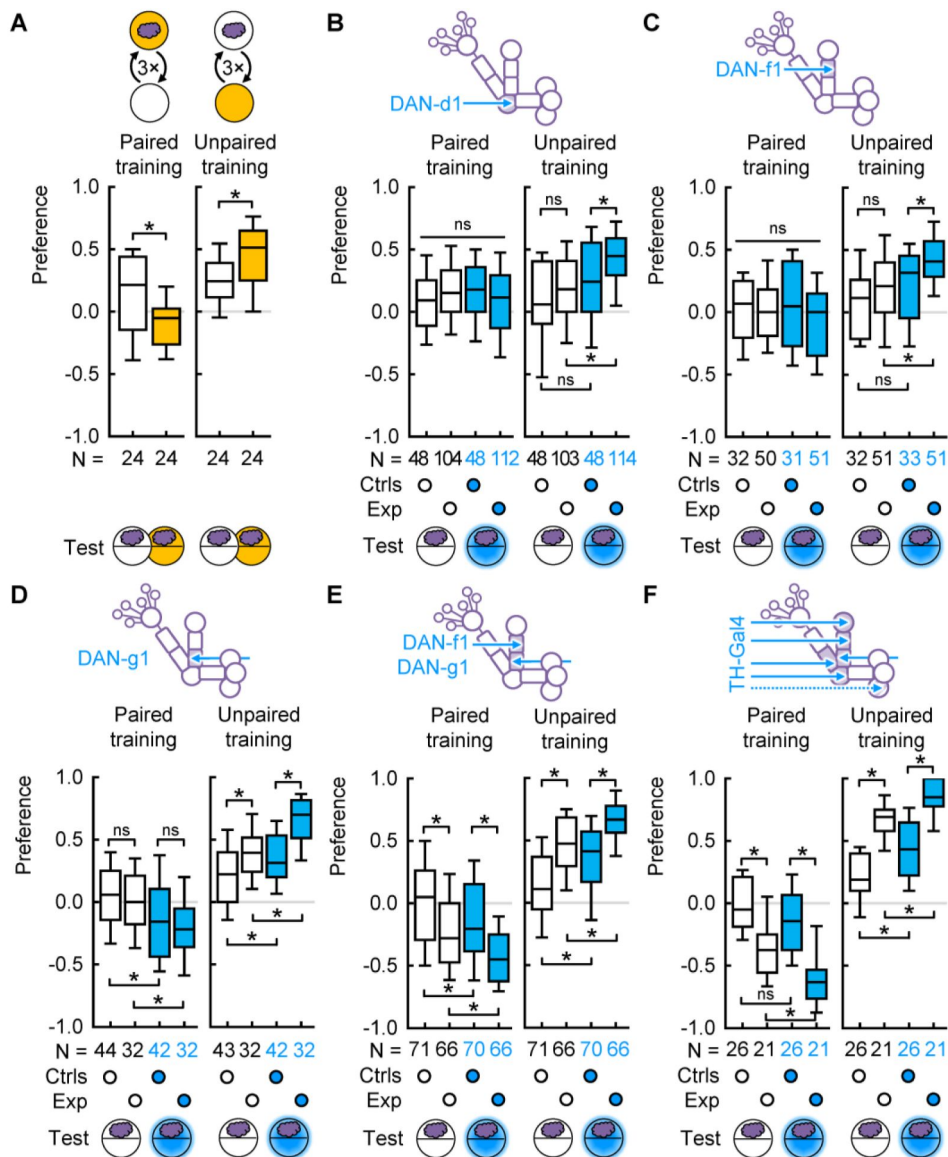


Figure 3

Individual DANs are sufficient to establish safety memory.

(A) With quinine as US, odour preferences after paired training were more negative in presence than in absence of the US (left, indicating punishment memory), and after unpaired training were more positive (right, indicating safety memory). (B) Left: after paired training, the animals of the pooled genetic controls and the experimental genotype (*DAN-d1>ChR2-XXL*) showed the same odour preference, both when tested in darkness or light. Right: after unpaired training, the experimental genotype showed a more positive odour preference than the genetic controls only when tested in light, indicating safety memory. (C) As in (B), but with *DAN-f1* activation. (D) Left: for *DAN-g1*, after paired training the animals of both the controls and the experimental genotype showed lower odour preferences when tested in presence than in absence of light. Right: after unpaired training, animals of the experimental genotype showed higher odour preferences than the controls, independent of the testing condition (right). (E) For a combination of *DAN-f1* and *DAN-g1*, the animals of the experimental genotype showed more negative odour preferences than the control after paired training (left, indicating punishment memory) and more positive preferences after unpaired training (right, indicating safety memory), independent of the testing condition. (F) For the *TH-Gal4-DANs*, similar results as in (E) were observed. ns above a line indicates non-significance across the whole data set (KW), * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

3D [↗](#), left). This suggests a potential punishment memory by the light, but not by the activation of DAN-g1. After unpaired training, the animals of the experimental genotype showed higher odour preferences than the controls, both in darkness and light (**Fig. 3D** [↗](#), right).

Taken together, we found that all three DANs induced only a safety memory after unpaired training, but no punishment memory after paired training (confirming similar results using *CsChrimson*¹⁵ [↗](#)). The safety memory established by DAN-d1 and DAN-f1 were retrieved only if the respective neuron was activated during the test, whereas the safety memory established by DAN-g1 was, at least partially, also retrieved without neuronal activation.

Combinations of DANs are sufficient for establishing both safety and punishment memories

We next wondered whether the activation of a combination of DANs would be sufficient to induce a punishment memory. For the combination of DAN-f1 and DAN-g1, after paired training, we indeed found more negative odour preferences in the experimental group than in the genetic controls, irrespective of the testing condition (**Fig. 3E** [↗](#), left). After unpaired training, we found higher odour preferences in the experimental genotype than in the genetic controls (**Fig. 3E** [↗](#), right). Finally, with *TH-Gal4* we observed analogous results (**Fig. 3F** [↗](#)).

Taken together, activating both DAN-f1 and DAN-g1, or activating most DANs in the brain, did establish both a punishment and a safety memory. Both kinds of memory were retrieved irrespective of whether the DANs were activated in the test or not.

Activating the DANs covered by TH-Gal4 modulates the animals' bending and velocity

Next, we investigated which effect activating the DL1-DANs would have on locomotion.

Using the novel IMBA (Individual Maggot Behaviour Analysis) software for behavioural analyses, we recently had found strong locomotor effects upon optogenetic activation of the DANs covered by *TH-Gal4*⁵¹ [↗](#). Within the current study, we repeated this experiment in total five times, in two different laboratories and with different parameters (**Fig. 4 – Figure supplement 1A** [↗](#)). This gave us the opportunity to probe the behavioural effects upon DAN activation for their robustness across experimental repetitions. Groups of around 8 experimentally naïve larvae of either the experimental genotype or the driver or effector controls were allowed to crawl freely on an agar-filled Petri dish in darkness. After 30 s, blue light was activated for 30 s, followed by darkness. We video-recorded the animals' behaviour and analysed it afterwards with the IMBA software (**Fig. 4A** [↗](#)).

The IMBA software provides a rich variety of behavioural attributes for individual animals. We selected 17 attributes (for a list and descriptions, see **table 1** [↗](#)) and first asked how these attributes relate to each other. Towards this end, we performed a Spearman correlation analysis of all 17 attributes based on the animals' baseline behaviour in darkness. We found three clusters of inter-correlated attributes (**Fig. 4B** [↗](#)). Cluster 1 consists of attributes measuring different aspects of directional changes, e.g. the animals' bending, or the rate of lateral head movements, the so-called head casts (HC). All attributes of cluster 2 relate to the speed of the larvae's peristaltic movement. Finally, cluster 3 consists of attributes dealing with the directional bias of the animals' behaviour. For example, the bending bias measures to which extent an animal was bent more towards one side than another.

Then, we wondered how much each attribute would change upon light stimulation. Towards this end, we calculated a Δ -value of each attribute as an individual's behaviour during the second half of the light stimulation minus its baseline behaviour during darkness (**Fig. 4A** [↗](#)). We used only

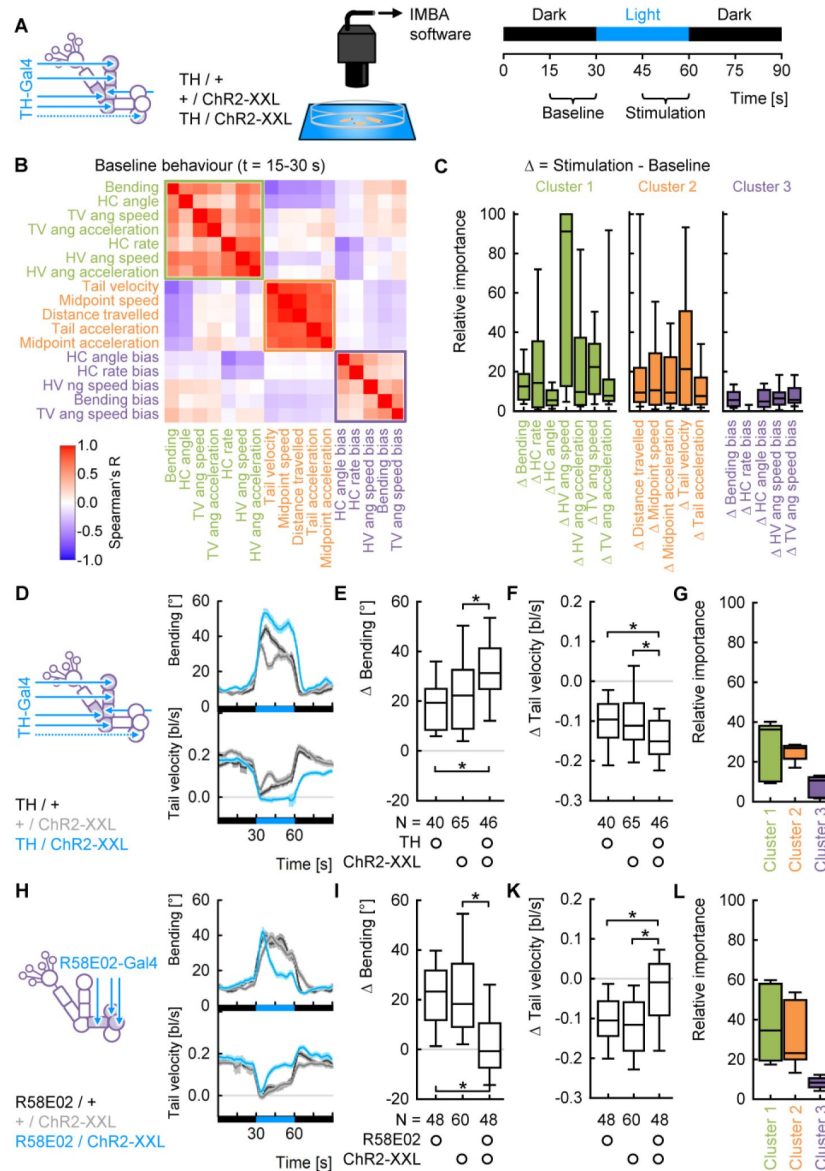


Figure 4

Two sets of DANs induce opposite effects on locomotion.

(A) Behaviour of cohorts of experimentally naïve larvae was recorded. After 30 s of darkness, blue light was activated for 30 s. The second halves of the dark and light periods are henceforth referred to as “baseline” and “stimulation”, respectively (B) Based on data from 5 experiments involving *TH>ChR2-XXL* and genetic controls within this study, the cross-correlation of 17 behavioural attributes during the baseline period revealed three clusters of inter-correlated attributes. Displayed is the Spearman's correlation coefficient *R* for each pair of attributes. Values of 1 (red) indicate perfect positive, values of -1 (blue) perfect negative correlations. (C) A random forest algorithm was trained to classify individuals of the experimental genotype and either genetic control, using their change in behaviour between the baseline and stimulation. Displayed is the relative importance of each behavioural attribute, scaled from 0 (least important) to 100 (most important). (D) Average bending and tail velocity over time. The black and blue bars at the bottom indicate the light regimen. (E) The change in bending upon light stimulation. (F) As in (E), but for the tail velocity. (G) The average relative importance of each attribute cluster, as provided by the random forest. (H-L) As in (D-G), but using *R58E02-Gal4* as driver. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

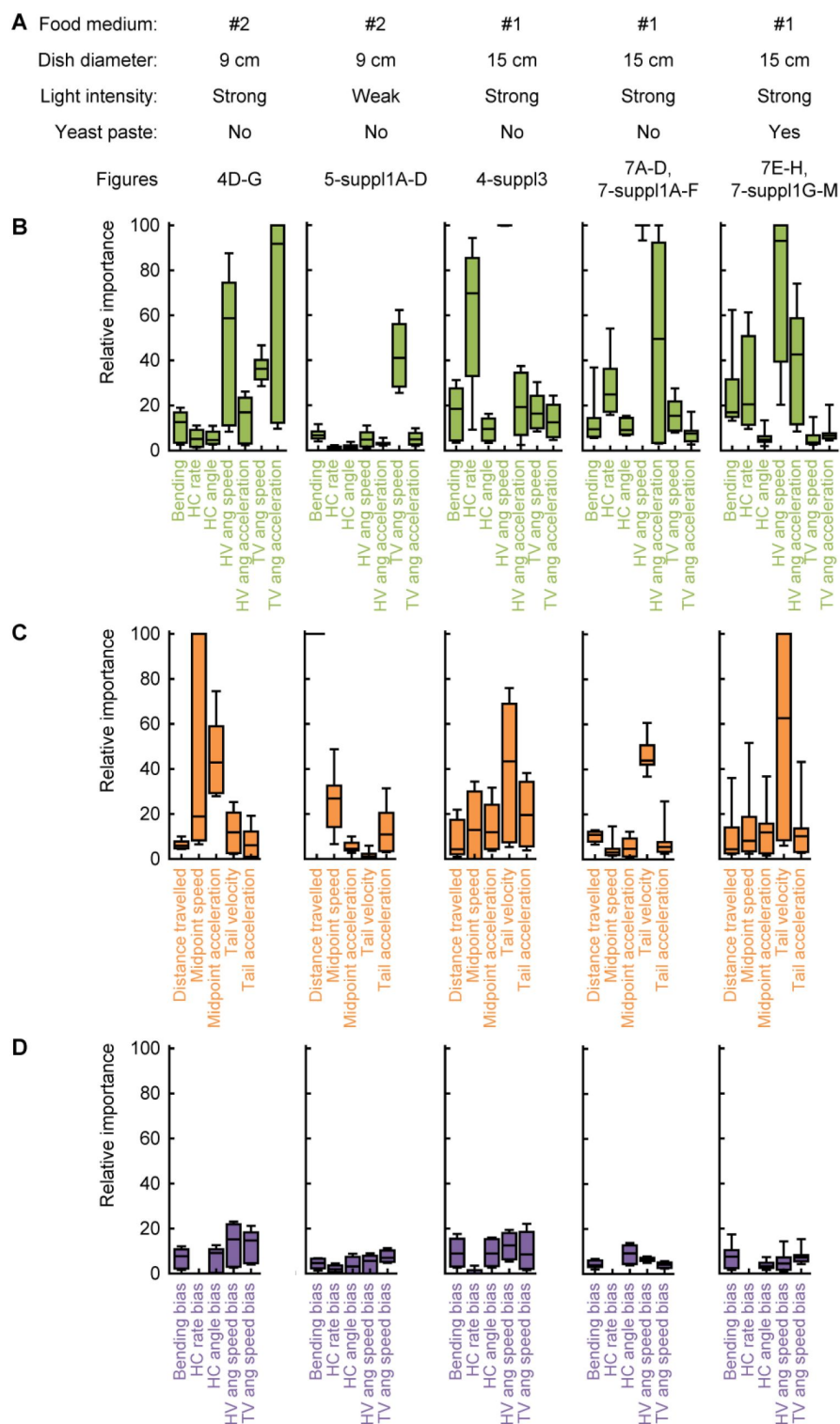


Figure 4 - supplement 1

Random forest analysis of 5 independent repetitions of the same locomotion experiment.

(A) Summary of the experimental conditions of five independent replications of the locomotion experiment with *TH-Gal4* and *ChR2-XXL*. For the right-most experiment, the larvae were placed on a yeast paste for 4 hours before the experiment (see Materials & Methods for details). (B-D) Each column shows the relative importance based on the random forest algorithm of a given experimental replication. (B), (C), (D) show the attributes of cluster 1, 2 and 3, respectively.

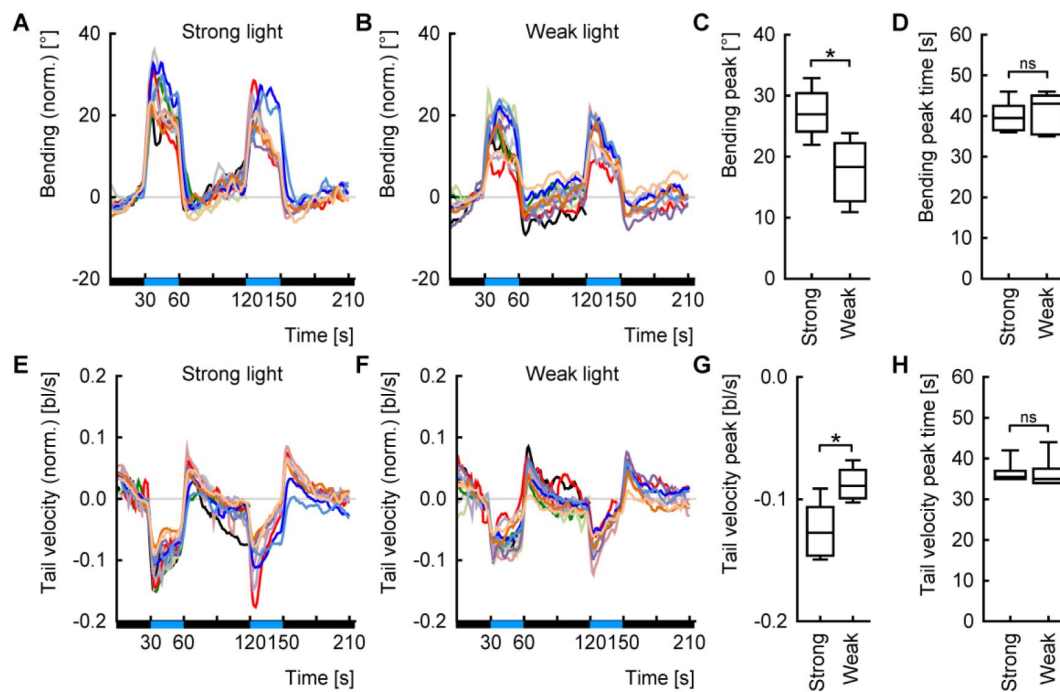


Figure 4 - supplement 2

Light responses of genetic controls.

(A) Bending over time. Black and blue bars at the x-axis indicate the light regimen. Each line represents the average bending of one genetic control of one of this study's experiment using strong blue light (specifically, the experiments shown in [Fig. 4](#), 8). The curve of each control was normalised to its bending during the time period between 15 and 30 s to allow comparisons of the shape of the curves independent from the baseline behaviour. Please note that some experiments included only one cycle of light stimulations (these curves end at time point 120 s), others included two cycles. (B) As in (A), but for experiments using weak blue light (using the experiments shown in [Fig. 5 – Figure supplement 1](#), [Fig. 8 – Figure supplement 2](#)). (C) The highest value during the first stimulation of each curve was determined. This peak value was higher for strong than for weak light. (D) The time until the peak of the first stimulation of each curve. No differences were seen between the light intensities. (E-H) As for (A-D), but for the tail velocity. As “peak” value, the lowest tail velocity of each curve during the first stimulation was used. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

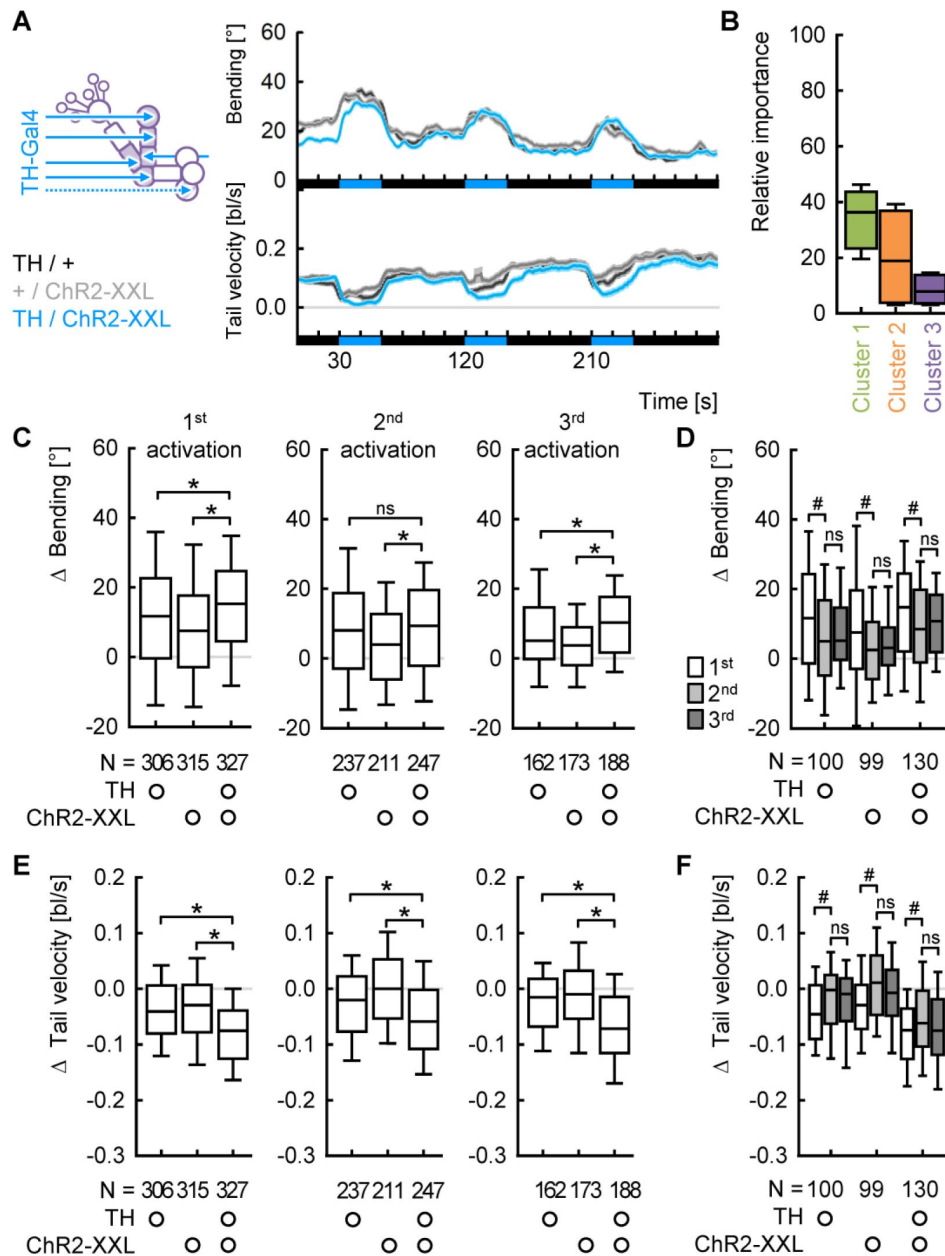


Figure 4 - supplement 3

Replication of the experiment shown in Fig. 4D-G with three stimulation cycles.

(A) Average bending and tail velocity over time. Larvae of all genotypes increased bending and decreased tail velocity upon light stimulation. Both effects were enhanced in the experimental genotype. The black and blue bars at the bottom indicate the light regimen. (B) The average relative importance of all behavioural attributes within each cluster, as provided by the random forest algorithm. (C) A Δ bending value was calculated in each individual to quantify the change in bending upon light stimulation. This change was significantly increased in the experimental genotype for each light stimulation. (D) As in (C), but only including individuals that were recorded during all three light stimulations. Differences across light stimulations are compared within individuals. The change in bending was reduced for the second and third stimulation in all genotypes. (E-G) As in (C-D), but for the tail velocity. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). # above brackets indicate pairwise within-animal significance (WSR). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

Name	Unit	Description	Cluster
Bending	°	The bending angle (BA) is the angle between the larva's tail vector and a vector from the middle of the larva to its head. Positive values indicate the larva bending to the left, negative values bending to the right. "Bending" in this study refers to the absolute, i.e. direction-independent BA.	1
Bending bias		The absolute average of all BA values of a larva, divided by the average of all absolute BA values: $\text{abs}(\overline{BA})/(\overline{\text{abs}(BA)})$. The Bending bias is constrained between 0 and 1, with 0 indicating a larva was bending equally to left and right, and 1 indicating a larva was only bending into one direction (no matter which direction).	3
HV angular speed	$\frac{^\circ}{s}$	The head vector angular speed (HVAS) is the angular speed of the head vector (HV) through the head of the larva. Positive values indicate movement of the head to the right, negative values to the left. "HV angular speed" in this study refers to the absolute, i.e. direction-independent HVAS.	1
HV angular speed bias		Calculated analogous to the Bending bias. Constrained between 0 and 1, with 0 indicating a larva was moving its head equally fast to left and right, and 1 indicating a larva was moving its head only into one direction.	3
HV angular acceleration	$\frac{^\circ}{s^2}$	The absolute angular acceleration of the head vector (HV).	1
TV angular speed	$\frac{^\circ}{s}$	The tail vector angular speed (TVAS) is the angular speed of the tail vector (TV) through the rear half of the larva. Positive values indicate movement to the right, negative values to the left. "TV angular speed" in this study refers to the absolute, i.e. direction-independent TVAS.	1
TV angular speed bias		Calculated analogous to the Bending bias. Constrained between 0 and 1, with 0 indicating a larva was moving its tail equally fast to left and right, and 1 indicating a larva was moving its tail only into one direction.	3
TV angular acceleration	$\frac{^\circ}{s^2}$	The absolute angular acceleration of the tail vector (TV).	1
HC rate	$\frac{HC}{s}$	The number of head casts (HC) per second. HCs were defined as events of a HVAS larger than 35 °/s or smaller than -35 °/s, respectively.	1
HC rate bias		The HC rate of HCs to the left minus the HC rate of HCs to the right, divided by the total HC rate. Constrained between 0 and 1, with 0 indicating a larva had equal HC rates to left and right, and 1 indicating a larva was making HCs in only one direction.	3
HC angle	°	The head cast angle (HCA) indicates how much the orientation of the animal changes upon each HC. It is defined as the difference between the bending angle at the end and the beginning of the HC. Positive values indicate that the larva changes its orientation to the left, negative values that it changes its orientation to the right. "HC angle" in this study refers to the absolute, i.e. direction-independent HCA.	1
HC angle bias		Calculated analogous to the Bending bias. Constrained between 0 and 1, with 0 indicating a larva was making equally large HCs to left and right, and 1 indicating a larva was making HCs in only into one direction.	3
Distance travelled	bl	The distance of the complete path a larva travels during the observation period (measured in body lengths [bl]).	2
Midpoint speed	$\frac{bl}{s}$	The speed of the larva's midpoint, independent of direction.	2
Midpoint acceleration	$\frac{bl}{s^2}$	The absolute acceleration of the larva's midpoint.	2
Tail velocity	$\frac{bl}{s}$	The velocity of the larva's tail point in a forward direction, that is in the direction of the tail vector.	2
Tail acceleration	$\frac{bl}{s^2}$	The absolute acceleration of the larva's tail in the direction of the tail vector.	2

Table 1

A list and description of all behavioural attributes used for locomotion analysis in this study.

the second half of the stimulation to exclude any potential startle-like response to the sudden light. Then, we used a random forest classification algorithm to classify individual larvae between the genotypes based on the 17 Δ -attributes. Crucially, the random forest provides the relative importance of each attribute for the classification. In other words, it can be used to obtain an unbiased measure of how relevant each attribute is for discriminating the behaviour of different genotypes⁵¹. We found consistently the attributes of cluster 1 and 2 to be much more important than those of cluster 3 (**Fig. 4C**). Within each cluster, most attributes were on average similarly important, with strong variability both within and across data sets (**Fig. 4 – Figure supplement 1**).

We concluded that the main behavioural differences between the experimental genotype and the genetic controls were the amount of bending and directional changes (cluster 1) as well as peristaltic speed (cluster 2). Given the correlations of individual attributes within each cluster (**Fig. 4B**) as well as their variable importance (**Fig. 4 – Figure supplement 1**), for all following analyses we decided to focus on one representative attribute of each cluster 1 (bending) and cluster 2 (tail velocity) that we have used already in a previous study⁵¹. In addition, we will present the relative importance of each attribute cluster along with each experiment.

Activating different subsets of DANs modulates locomotion in opposite directions

In accordance with our previous results⁵¹, we found that the control genotypes increased their bending and decreased their tail velocity upon light stimulation (**Fig. 4D**, Supplementary Movie 2). Such responses to blue light have been reported before⁵² and were consistently seen throughout this study (**Fig. 4 – Figure supplement 2**). Both types of behavioural change were significantly enhanced in the experimental genotype in which the *TH-Gal4*-DANs were activated⁵¹ (**Fig. 4D-F**). More specifically, when the DANs were activated, the larvae kept high bending and low velocity throughout the 30 seconds of light stimulation whereas the controls gradually reverted towards their normal behaviour (**Fig. 4D**). These results were confirmed by a repetition of the experiment using larger Petri dishes and multiple light stimulations (**Fig. 4 – Figure supplement 3**). For the random forest, the attributes from cluster 1 and 2 were equally important (**Fig. 4G**, **Fig. 4 – Figure supplement 3**).

As *TH-Gal4* covers all DANs except those of the pPAM cluster, we next asked whether activation of the pPAM DANs would have a similar effect on locomotion. Towards this end, we used *R58E02-Gal4* which covers 3 out of 4 pPAM DANs (DAN-h1, DAN-i1 and DAN-j1)²⁷. To our surprise, activation of these DANs decreased bending and increased tail velocity compared to the controls (**Fig. 4H-K**, Supplementary Movie 3). More precisely, after an initial reaction to switching on the light, when the pPAM DANs were activated the larvae reverted to their normal behaviour much quicker than the controls (**Fig. 4H**). Again, the attributes of cluster 1 and 2 were similarly important for the random forest classification (**Fig. 4L**).

Thus, the activation of two non-overlapping sets of DANs had opposite effects on behaviour: activating the *TH-Gal4*-DANs (including the aversive US-signalling DANs) increased bending and decreased velocity, whereas activating the *R58E02-Gal4*-DANs (which were shown to signal appetitive US)^{27,47,53} decreased bending and increased velocity.

DANs modulate locomotion independent from the light responses

In the previous experiments, animals of all genotypes displayed a very strong response to the light stimulation. We therefore wondered whether the observed effects upon DAN activation were due to a modulation of this light response, or an independent modulation of the animals' locomotion.

We compared the animals' behavioural changes in three sets of experiments, activating the DANs with strong blue light, weak blue light, or red light, respectively (**Fig. 5**). When using the about 10-fold lower light intensity, we nevertheless observed rather strong light responses in the controls (**Fig. 4 – Figure supplement 2**), and robust differences in behaviour upon activation of either set of DANs (**Fig. 5 – Figure supplement 1**). These effects gradually diminished with prolonged light stimulation (**Fig. 5 – Figure supplement 2**).

Red light reportedly triggers only very mild light responses^{16,52}. We therefore next used the red-shifted effector CsChrimson⁵⁴, and raised larvae either with or without all-trans-retinal which is necessary for the Chrimson channels to function⁵⁴ (**Fig. 5 – Figure supplement 3**). When the larvae were raised with retinal, we observed similar effects as when using ChR2-XXL (**Fig. 5 – Figure supplement 3**). The control without retinal, in contrast, displayed only very weak responses to the red light (**Fig. 5 – Figure supplement 4**).

In order to compare the effects of DAN activation between the three experiments, we defined the behaviour of the pooled controls in each experiment as zero, and displayed the difference of each experimental group from their respective controls. The behavioural effects of the DAN activations were strikingly similar in the three conditions, both in magnitude and time course (**Fig. 5A-B**). The only obvious difference was an extended reduction of the tail velocity beyond the activation of the *TH-Gal4*-DANs that could be seen only when using ChR2-XXL (**Fig. 5A**). This “after-effect” may be due to the reportedly slow closing times of the ChR2-XXL channel⁴¹.

We wanted to further quantify the relation between the light responses elicited in each experiment and the behavioural effects of the DAN activation. Therefore, we used the mean behavioural changes upon light stimulation of the pooled controls of each experiment as an indicator for the effect size of the light response, and the difference in behavioural changes between the experimental groups and the respective controls as an indicator for the effect size of the DAN activation. We found that the effect sizes of the DAN activation were all relatively similar and completely unrelated to the effect sizes of the light responses (**Fig. 5C**).

In summary, we concluded that the observed modulations of bending and tail velocity upon DAN activation occurred independent of the animals' responses to the light stimulation itself.

Different subsets of DANs convey opposite acute valence when activated

Next, we asked how to interpret the behavioural effects of activating DANs. Stopping and turning around upon a sudden activation of *TH-Gal4*-DANs can be interpreted as a reaction to an aversive signal^{16,52} – on the other hand, slowing down and turning on the spot could also be a behavioural strategy to stay in a pleasant situation. Likewise, going straight and speeding up upon *R58E02-Gal4*-DAN activation can be interpreted as a “go on”-reaction to an appetitive signal or a strategy to escape. To test for the valence of activation of either set of DANs we performed choice experiments between a dark and an illuminated half of a Petri dish (**Fig. 6A**). Larvae of all genotypes avoided the illuminated half as expected^{42,52}. Relative to the genetic controls, larvae with *TH-Gal4*-DANs activated showed more avoidance, and larvae with *R58E02-Gal4*-DANs showed more approach (**Fig. 6B-C**, Supplementary Movies 4-5). We next analysed the animals' behaviour near the dark-light-border. When activating *TH-Gal4*-DANs, the animals' bending was increased and their velocity decreased on the illuminated half compared to the controls (**Fig. 6D-F**), similar to our previous experiments (**Fig. 4**). When activating *R58E02-Gal4*-DANs, the animals' bending was decreased and their velocity increased compared to the controls everywhere on the dish (**Fig. 6G-I**). Taken together, we concluded that activating *TH-Gal4*-DANs conveys an acute aversive signal and *R58E02-Gal4*-DANs conveys an acute appetitive signal. The animals' reaction to the sudden activation of these DANs therefore should be interpreted as aversive and appetitive behaviour, respectively.

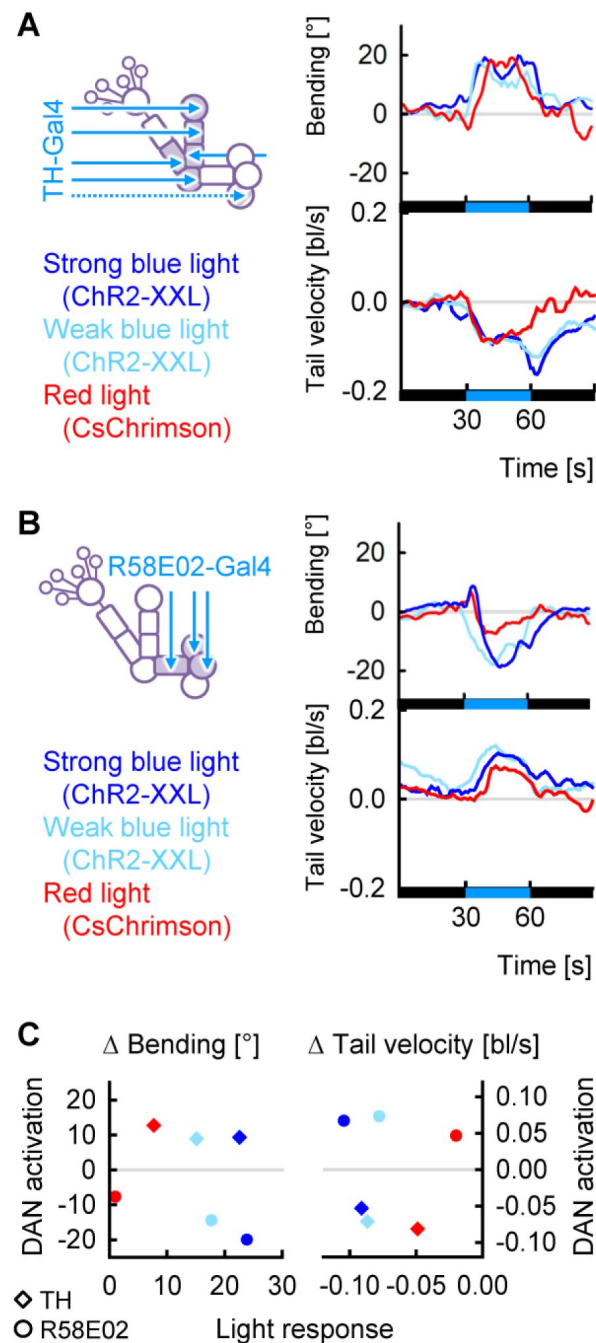


Figure 5

DAN activations modulate locomotion independent of light.

(A) Average difference in bending and tail velocity of the experimental group from the control groups over time. Black and blue bars at the x-axis indicate the light regimen. Each curve represents the behaviour of the experimental group minus the pooled controls of a given experiment to indicate the effect of DAN activation, “purified” from the light responses. Compared are experiments using strong blue light (dark blue, **Fig. 4D**) and weak blue light (light blue, Figure supplement 1A) with ChR2-XXL as effector, and red light (red, Figure supplement 3A) with CsChrimson. (B) As in (A), but for the experiments shown in **Figures 4H**, Figure supplement 1E and 3E. (C) For each of the 6 experiments, the average Δ bending (left) or Δ tail velocity (right) of the pooled control groups was taken as the effect size of the light response (x-axis). The Δ bending or Δ tail velocity of the experimental group minus those of the pooled control groups was taken as the effect size of DAN activation (y-axis). Note that the left and right graphs use different axes and scaling. The colours match those of (A-B). All experiments using the same driver resulted in similar effect sizes of DAN activation, despite strong differences in effect sizes of the light response.

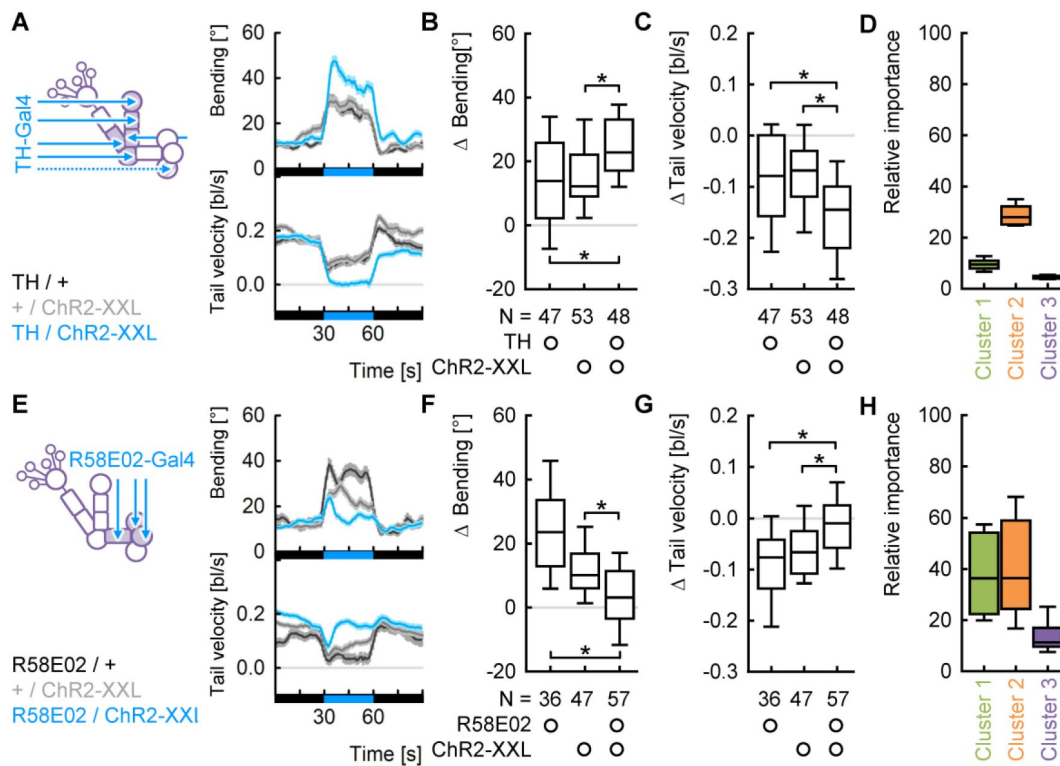


Figure 5 - supplement 1

DAN activations modulate locomotion independently of light intensity.

Replication of the experiments and analyses of the experiments displayed in **Fig. 4D-L**, but using weak blue light during the stimulations. (A,E) Average bending and tail velocity over time. (B,F) Change in bending upon light stimulation. (C,G) Change in tail velocity upon light stimulation. (D,H) Average relative importance of the behavioural attributes within each cluster, as provided by the random forest algorithm. For *TH-Gal4*, cluster 1 was less important than in the case of strong light, but the change in bending was nevertheless significant. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

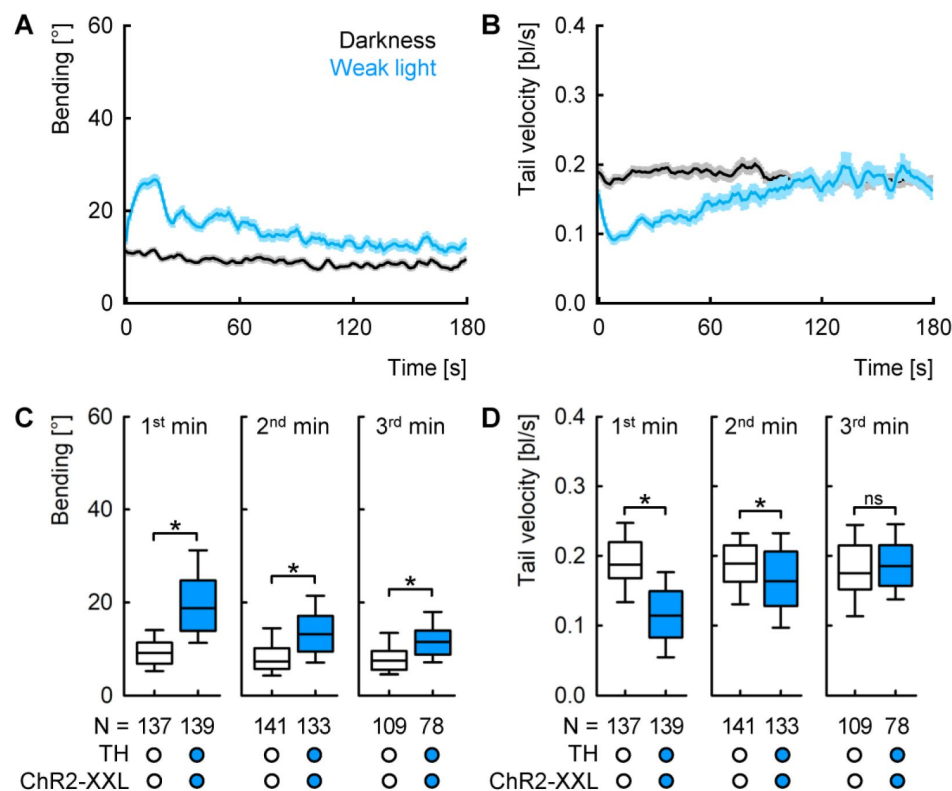


Figure 5 - supplement 2

DAN activation during innate odour preference experiments causes increased bending and decreased velocity.

(A) Larvae were video-recorded and their behaviour analysed during the experiment displayed in **Fig. 2 – Figure supplement 1F**. The larvae of the experimental genotype (*TH-Gal4>ChR2-XXL*) had a higher average bending in blue light than in darkness. (B) The larvae showed a lower tail velocity in light than in darkness during the first half of the experiment. Both differences were much weaker than in the experiments presented in Figure supplement 1 and diminished with time. (C) Quantification of (A) on the basis of individual animals. From left to right, the average bending during the first, second and third minute was examined. (D) as in (B), but for tail velocity. In summary, with the experimental parameters of the first half of this study, the animals' locomotion was only altered moderately, in a way that still allowed them to reach their goal. A possible reason for the differently strong light response may be the different handling of the animals: in the learning and preference experiments, the Petri dish was loaded with the larvae in dimmed room light and placed in the experimental setup, followed by an immediate start of the light stimulation and video recording. In the locomotion experiments, the larvae were kept in darkness for at least 40 s before the light stimulation started. The sudden light stimulus after a period of darkness may trigger stronger light responses. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

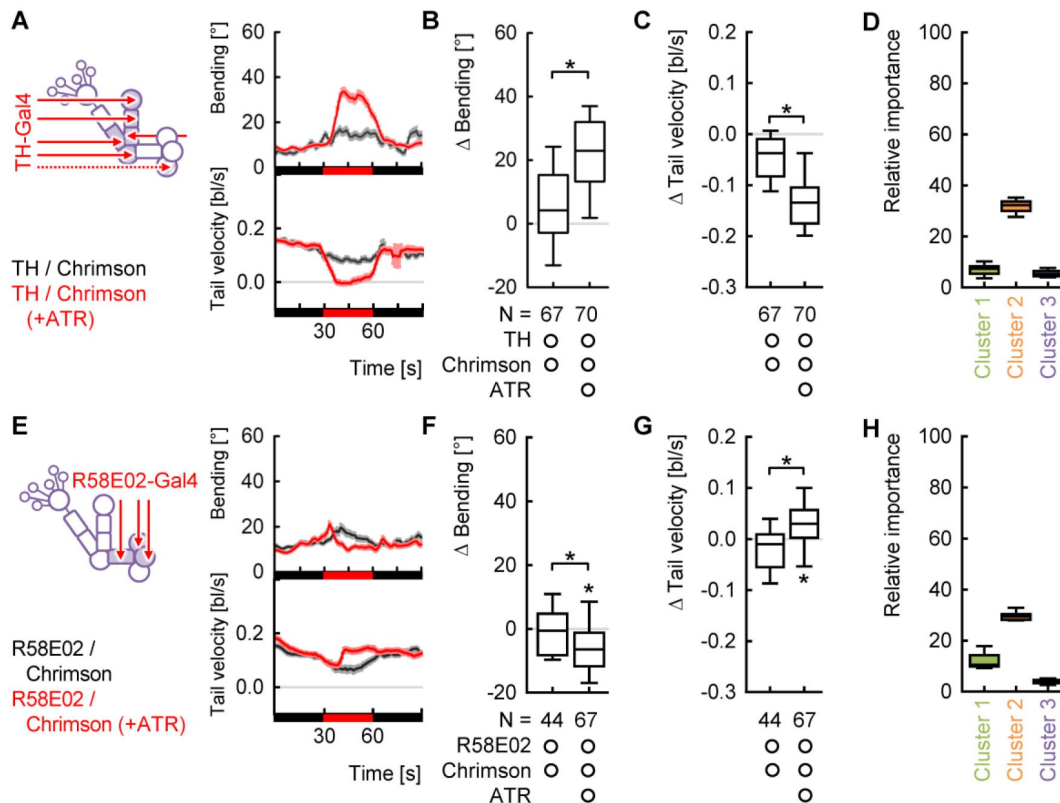


Figure 5 - supplement 3

DAN activations modulate locomotion independently of effector.

Replication of the experiments and analyses of the experiments displayed in **Fig. 4D-L**, but using CsChrimson as effector and red light stimulation. The experimental groups were larvae expressing CsChrimson in *TH-Gal4*-DANs (A-D) or *R58E02-Gal4*-DANs (E-H), fed with all-trans-retinal. The control groups were of the same genotype in standard food without all-trans-retinal. Note that these control animals behaved comparably to an effector control (Figure supplement 4), indicating that the DANs were not activated in the controls. (A,E) Average bending and tail velocity over time. (B,F) Change in bending upon light stimulation (C,G) Change in tail velocity upon light stimulation. (D,H) Average relative importance of the behavioural attributes within each cluster, as provided by the random forest algorithm. Cluster 1 was less important than in the case of blue light, but the changes in bending was nevertheless significant. Strikingly, activating *R58E02-Gal4*-DANs decreased bending below baseline and increased velocity above baseline rather than just reverting the behaviour back to baseline. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

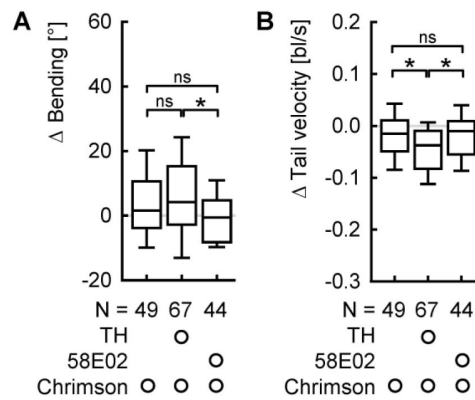


Figure 5 - supplement 4

Comparison of controls for the experiments shown in Figure supplement 3.

The behaviour of larvae expressing CsChrimson either (A) under the control of *TH-Gal4* or (B) *R58E02-Gal4* were compared with the heterozygous effector control. None of the genotypes were raised with all-trans retinal. Overall, the controls showed almost no reaction to the light. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

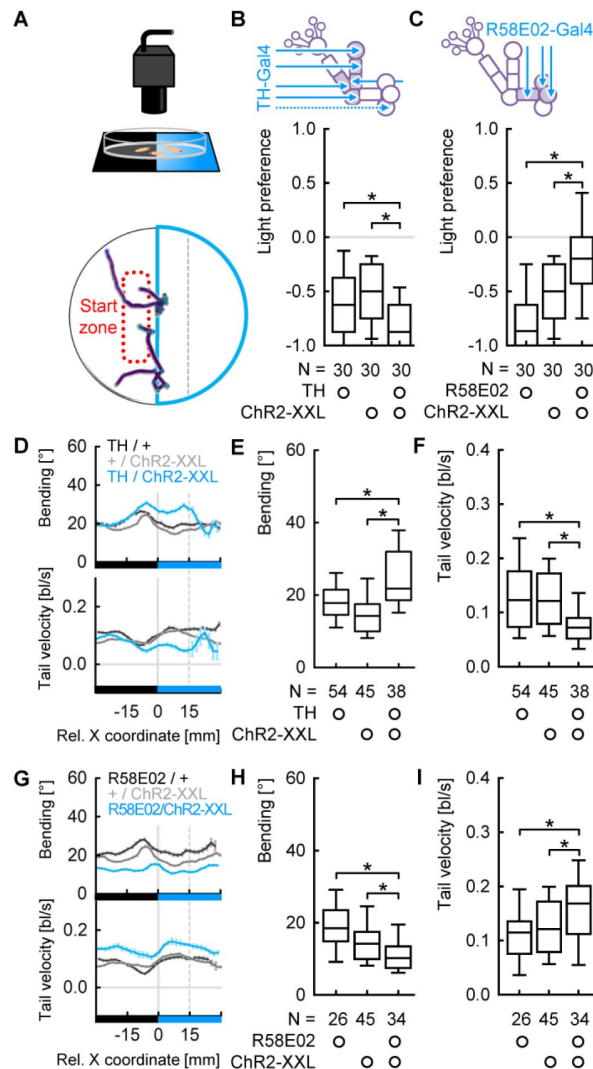


Figure 6

Different subsets of DANs convey opposite acute valence when activated.

(A) Larvae were given the choice between a dark and an illuminated (weak blue light) half of a Petri dish and their behaviour was recorded for three minutes. Animals were placed into the dark half at the dish. Displayed are 2 sample tracks for animals of *TH>ChR2-XXL*. (B) The light preference after 3 minutes. All genotypes showed light avoidance which was enhanced in the experimental genotype. (C) As in (B), but for *R58E02-Gal4* as driver. The experimental genotype showed less light avoidance than the controls. (D) Average bending and tail velocity of *TH>ChR2-XXL* and genetic controls over the X coordinate of the Petri dish, with 0 indicating the midline. The black and blue bars at the bottom indicate which half was illuminated. (E) The average bending on the illuminated side of the dish. We limited the analysis to an area up to 15 mm from the midline (indicated as stippled line in A) to capture the behaviour in the choice zone. (F) As in (E), but for the tail velocity. (G-I) As in (D-F), but for *R58E02-Gal4* as driver. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

The modulations of bending and velocity are dopamine-dependent

Several DANs have been found to have additional neurotransmitters^{19,55}. Therefore, we wondered whether the observed effects on locomotion were indeed dependent on dopamine signalling. Towards this end, we combined optogenetic activation by ChR2-XXL with TH-RNAi which knocks down the expression of tyrosine hydroxylase, the rate limiting enzyme of dopamine synthesis. Thus, when both effectors are expressed in a set of DANs, these DANs can get activated by light but have a reduced dopamine synthesis. When expressing both effectors in the *TH-Gal4*-DANs, we found that the change in bending upon DAN activation was slightly decreased compared to animals without the TH-RNAi, but still higher than in the genetic controls (**Fig. 7A-B**, **Fig. 7 – Figure supplement 1A-F**). Furthermore, in animals with TH-RNAi the tail velocity was generally increased, but the change in tail velocity upon DAN activation was unaffected (**Fig. 7A,C**, **Fig. 7 – Figure supplement 1A-F**). There are several possible reasons for this lack of effects, including a too weak knock-down of tyrosine hydroxylase or a developmental adaptation to the reduced dopamine levels. Therefore, we next sought to inhibit tyrosine hydroxylase by an independent and acute technique.

Larvae were fed the tyrosine hydroxylase inhibitor 3IY for four hours before the experiment, following established protocols^{56,57}. 3IY-fed larvae showed a change in bending upon DAN activation that was significantly lower than control animals without 3IY and on the same level as genetic controls (**Fig. 7E-F**, **Fig. 7 – Figure supplement 1G-M**). Interestingly, the baseline tail velocity was decreased rather than increased compared to animals without 3IY feeding (**Fig. 7E**, **Fig. 7 – Figure supplement 1K**). This means inhibiting the activity of tyrosine hydroxylase acutely or during the whole development had opposite effects on the baseline tail velocity. Upon DAN activation, 3IY-fed animals reduced their tail velocity less than animals without 3IY, and to a level comparable to the genetic controls (**Fig. 7E,G**, **Fig. 7 – Figure supplement 1G-M**). These effects, both regarding the changes in bending and tail velocity as well as the shifted baseline velocity, were replicated in two independent experiments (**Fig. 7 – Figure supplement 2**).

Taken together, we concluded that the locomotor effects upon activation of the *TH-Gal4*-DANs were largely dopamine-dependent, at least when inhibiting the tyrosine hydroxylase acutely with 3IY.

Individual DANs are sufficient to increase bending and decrease velocity

Given the broad expression pattern of *TH-Gal4*, we wondered which cells were responsible for the observed modulations of bending and velocity. Therefore, we probed the DANs of the DL1 cluster individually with experiments that included two cycles of light stimulation (**Fig. 8 – Figure supplement 1**). Overall, we found similar differences between the experimental genotype and the controls upon both stimulations, but more enhanced during the second stimulation.

For DAN-d1, we found that the experimental genotype displayed an increased bending and decreased tail velocity compared to the controls that remained trends for the first stimulation (**Fig. 8 – Figure supplement 1A-C**) and were statistically significant for the second stimulation (**Fig. 8A-D**). The effect on bending seemed to be somewhat weaker than for *TH-Gal4* and expressed itself in a delayed peak rather than a general increase. The tail velocity, in contrast, was reduced to zero in both cases (compare **Fig. 4D-G** and **8A-D**). However, using weaker light did not result in any significant differences between the experimental genotype and the control, in contrast to the case of *TH-Gal4* (**Fig. 8 – Figure supplement 2A-C**). Thus, activating DAN-d1 individually has similar effects as activating all neurons covered by *TH-Gal4*, but two stimulations of strong light were required to fully induce these effects.

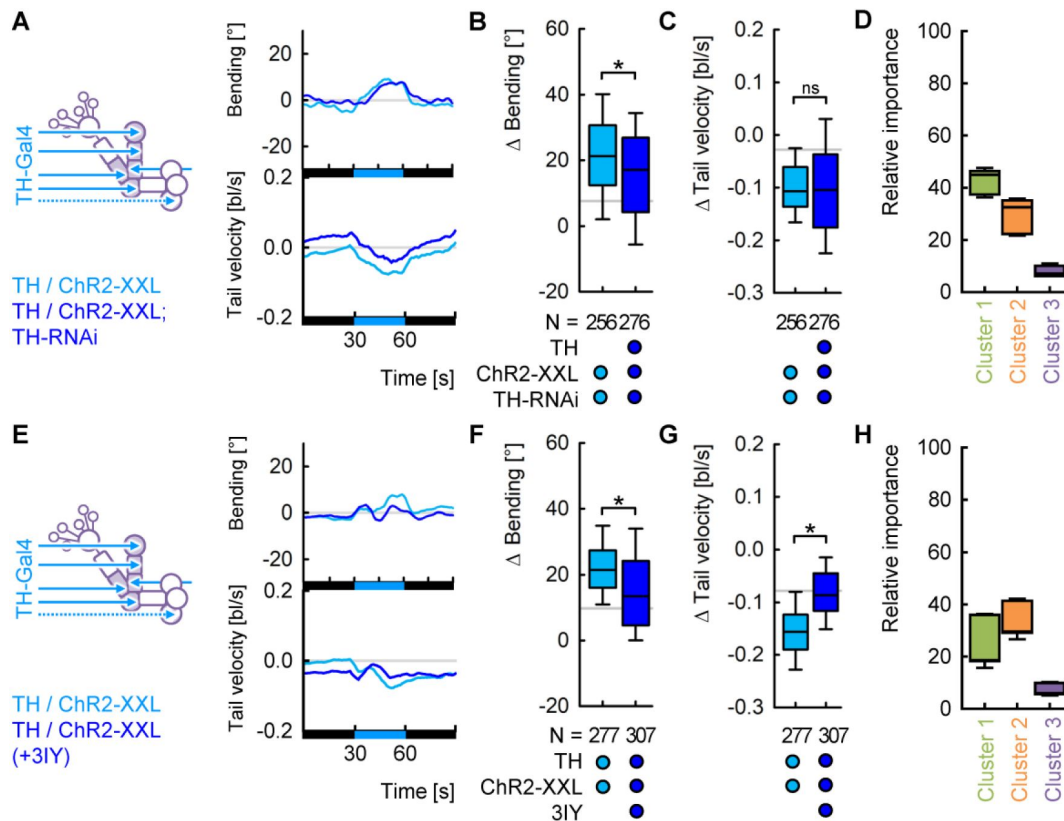


Figure 7

The effects on locomotion upon DAN activation are dopamine-dependent.

(A) Average difference in bending and tail velocity of the experimental groups from the control groups over time. Black and blue bars at the x-axis indicate the light regimen. Each curve represents the behaviour of an experimental group minus the pooled controls to indicate the effect of DAN activation, “purified” from the light responses. Compared are larvae expressing ChR2-XXL in *TH-Gal4*-DANs with or without additional *TH-RNAi*. (B) Change in bending upon DAN activation. The grey line indicates the median of the pooled genetic controls. For the data of all genotypes, see Figure supplement 1. (C) As in (B), but for tail velocity. (D) The average relative importance of each attribute cluster for a comparison between animals with and without *TH-RNAi*. (E-H) As in (A-D), but for animals of the experimental genotype that were or were not fed the TH inhibitor 3IY. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

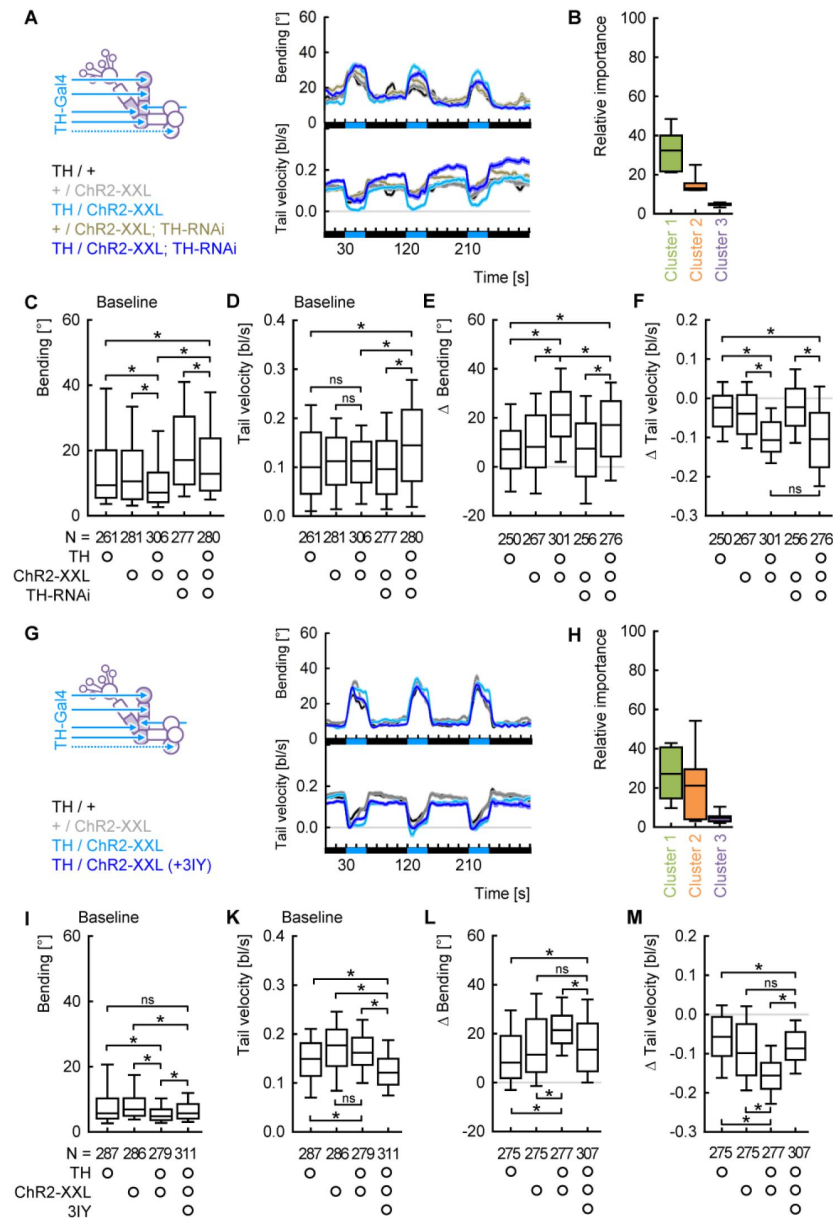


Figure 7 - supplement 1

The effects on locomotion upon DAN activation are dopamine-dependent.

(A) Average bending and tail velocity over time. The black and blue bars at the bottom indicate the light regimen. (B) The average relative importance of all behavioural attributes within each cluster, as provided by the random forest algorithm when comparing the experimental genotype with the genetic controls. (C) The animals' average bending during the baseline period (15 to 30 s) in darkness. Both genotypes including the *UAS-TH-RNAi* construct displayed increased bending, suggesting a genetic background effect. (D) As for (C), but for the tail velocity. Animals expressing *ple-dsRNA* (TH-RNAi) had an increased velocity compared to all other genotypes. (E) The change in bending upon the first light stimulation. This change was significantly increased in the experimental genotype and slightly reduced when in addition expressing *ple-dsRNA*. (F) As in (E), but for the tail velocity. (G-M) As for (A-F), but feeding the TH inhibitor 3IY instead of inducing *TH-RNAi*. The changes in both bending and tail velocity upon light stimulation for animals fed with 3IY were statistically not different from the genetic controls. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

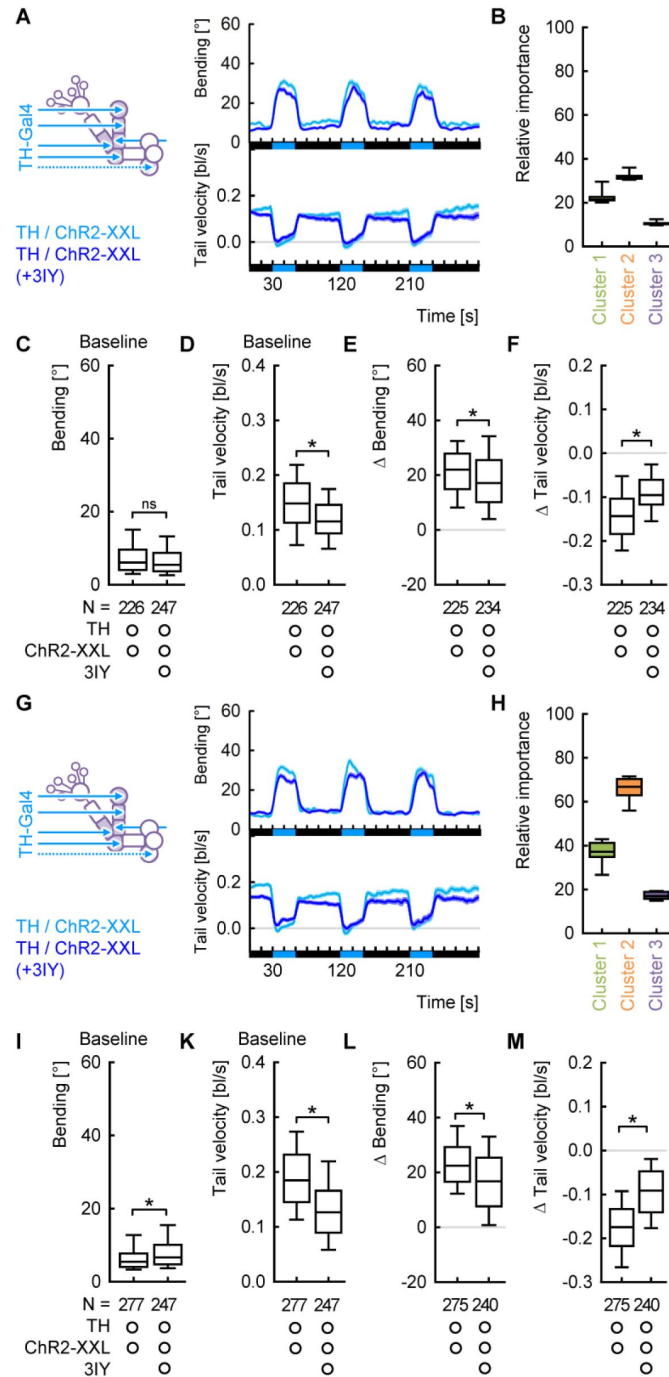


Figure 7 - supplement 2

Two independent replications of the experiments shown in Figure supplement 1.

(A,G) Average bending and tail velocity over time. The black and blue bars at the bottom indicate the light regimen. (B,H) The average relative importance of all behavioural attributes within each cluster, as provided by the Random forest algorithm. (C,I) The animals' average bending during the baseline period (15 to 30 s) in darkness. Feeding of 3IY had no effect. (D,K) As for (C,I), but for the tail velocity. Feeding 3IY reduced the tail velocity. (E,L) The change in bending upon the first light stimulation. This change was significantly decreased in animals fed with 3IY. (F,M) As in (E,L), but for the tail velocity. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

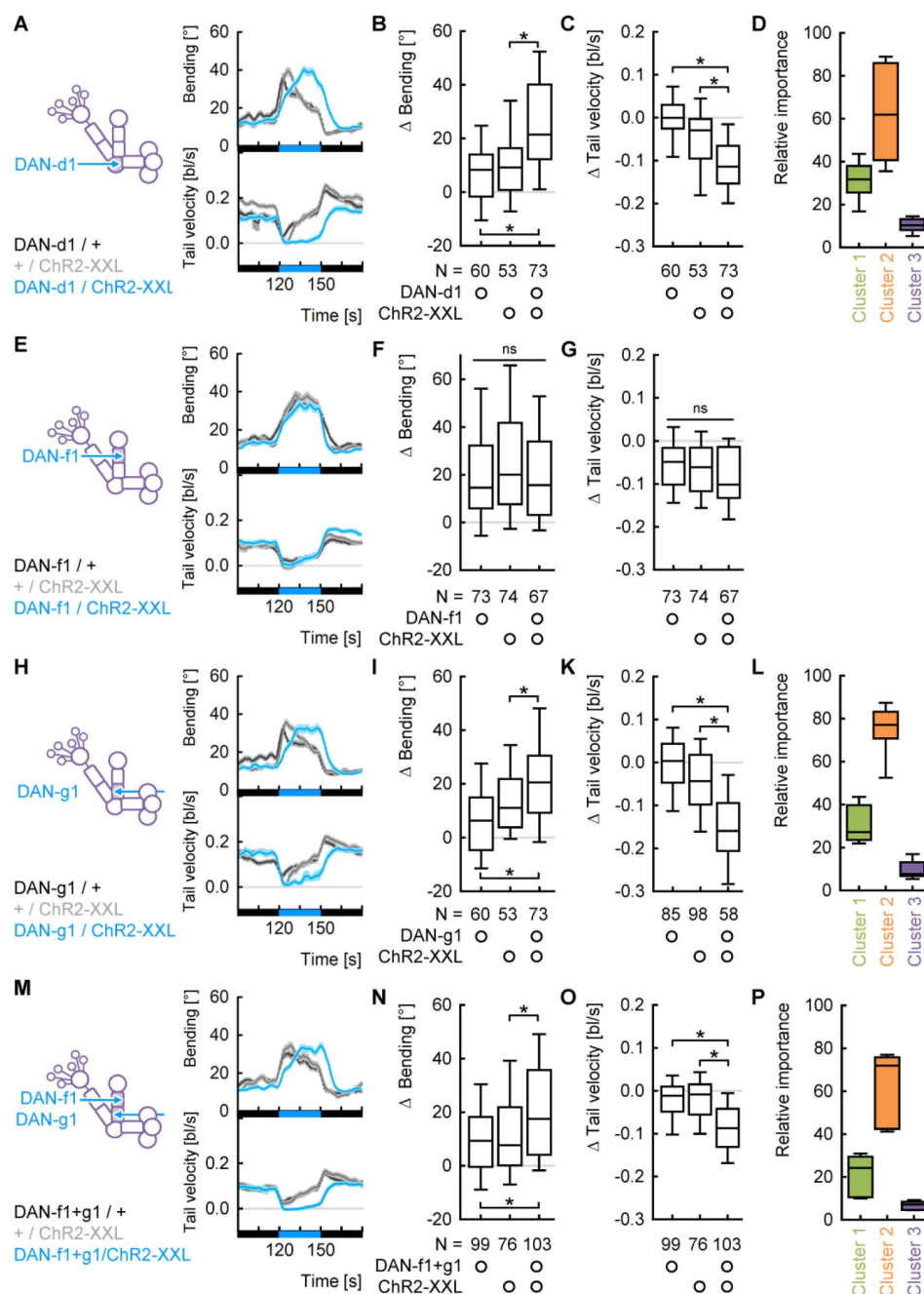


Figure 8

Individual DANs are sufficient to modulate bending and tail velocity.

(A) Average bending and tail velocity over time. The black and blue bars at the bottom indicate the light regimen. Note that the second light stimulation is shown. For the full time course and a quantification of the first stimulation, see Figure supplement 1. (B) The change in bending upon the second light stimulation was significantly increased when DAN-d1 was activated. (C) As in (B), but for the tail velocity. (D) The average relative importance of each attribute cluster, as provided by the random forest algorithm. Cluster 2 was clearly most important. (E-G) As in (A-C), but for DAN-f1. As no effects of DAN-f1 activation were seen, no random forest was applied. (H-L) As in (A-D), but for DAN-g1. (M-P) As in (A-D), but for a combination of DAN-f1 and DAN-g1. ns above a line indicates non-significance across the whole data set (KW), * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

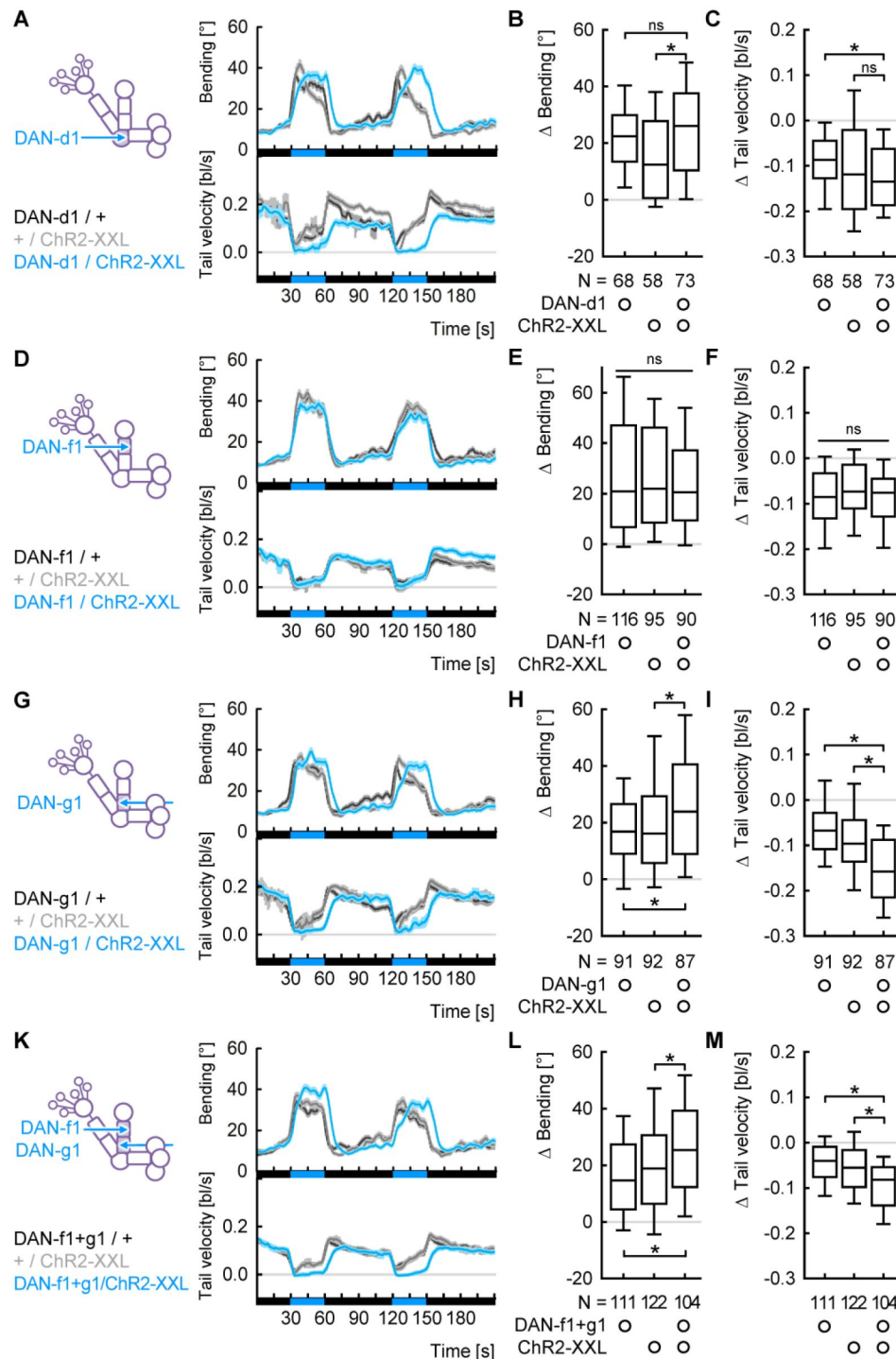


Figure 8 - supplement 1

Individual DANs are sufficient to modulate bending and tail velocity.

(A) Average bending and tail velocity over time. Notably, the difference between controls and experimental genotype (*DAN-d1>ChR2-XXL*) was more pronounced at the second stimulation. Partially, this was due to a more transient light response of the controls. The black and blue bars at the bottom indicate the light regimen. (B) The change in bending upon the first light stimulation. (C) As in (B), but for the tail velocity. (D-F) As for (A-C), but for *DAN-f1*. (G-I) As for (A-C) but for *DAN-g1*. (K-M) As for (A-C) but for a combination of *DAN-f1* and *DAN-g1*. ns above a line indicates non-significance across the whole data set (KW), * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

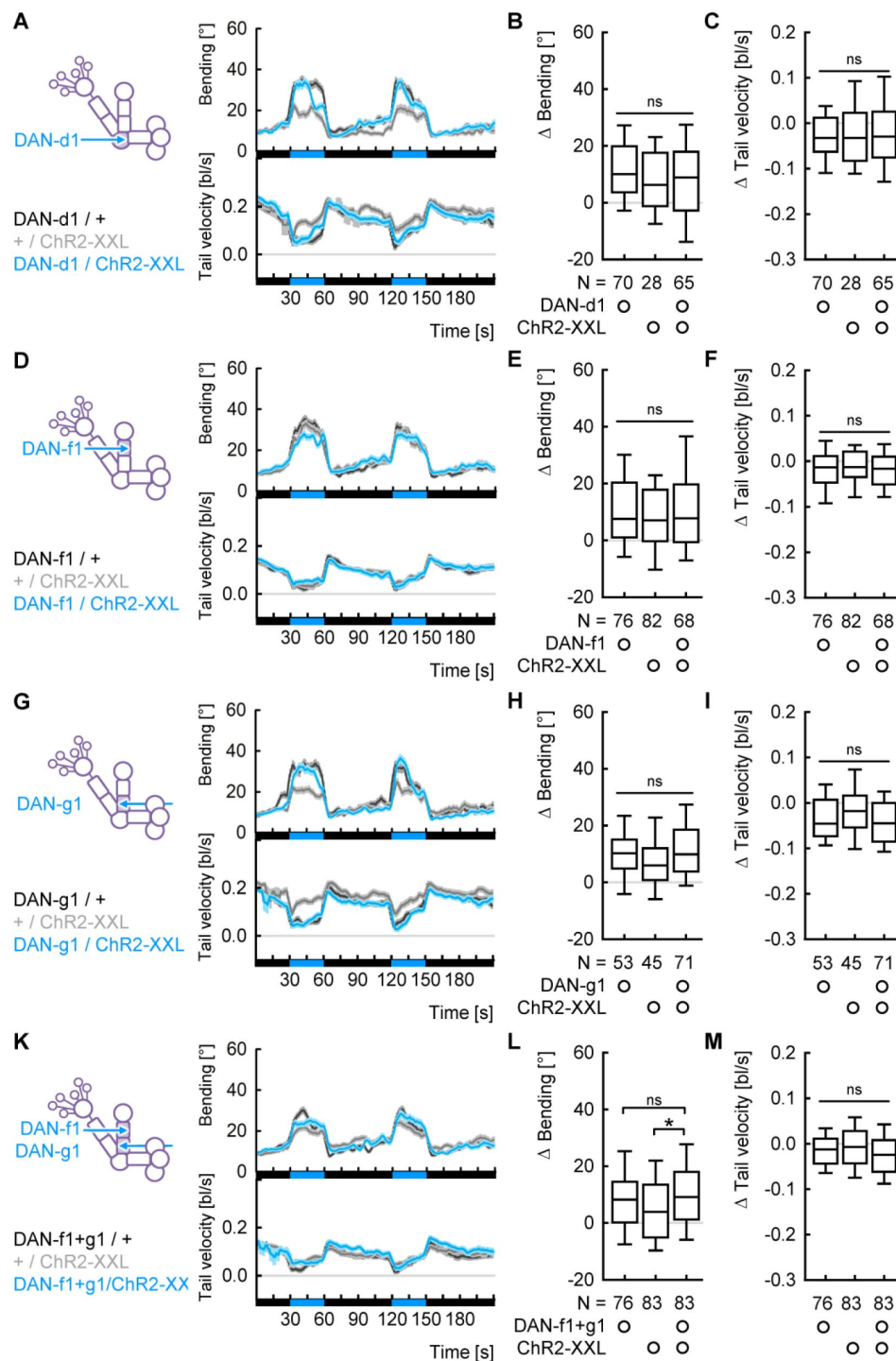


Figure 8 - supplement 2

Activation of individual DANs with weak light is not sufficient to modulate locomotion.

(A) Average bending and tail velocity over time. The black and blue bars at the bottom indicate the light regimen. (B) The change in bending upon light stimulation. The change was unchanged by activating DAN-d1 with weak light. (C) As in (B), but for the tail velocity. (D-F) As for (A-C), but for DAN-f1. (G-I) As for (A-C) but for DAN-g1. (K-M) As for (A-C) but for a combination of DAN-f1 and DAN-g1. ns above a line indicates non-significance across the whole data set (KW), * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data S1.

Next, we repeated the same experiments for DAN-f1 and DAN-g1. We found no effects on locomotion upon activating DAN-f1 (**Fig. 8E-G**, **Fig. 8 – Figure supplement 1D-F**, **Fig. 8 – Figure supplement 2D-F**). For DAN-g1, we found both significantly increased bending and decreased velocity very similar to DAN-d1, again only for strong light and clearer during the second activation, just like for DAN-d1 (**Fig. 8H-L**, **Fig. 8 – Figure supplement 1G-I**, **Fig. 8 – Figure supplement 2G-I**). Finally, we activated DAN-f1 and DAN-g1 simultaneously and found increased bending and decreased velocity comparable to the activation of DAN-g1 alone (**Fig. 8M-P**, **Fig. 8 – Figure supplement 1K-M**, **Fig. 8 – Figure supplement 2K-M**).

Taken together, our experiments demonstrated that two of the three tested DANs, when activated individually, induced qualitatively similar effects on bending and velocity that also the combination of almost all DANs induced.

Discussion

In this study, we found that the optogenetic activation of the same individual DANs can have multiple effects in establishing associative memories, retrieving memories and acutely modulating innate locomotion (**Fig. 9A**). How can such diverse functions be integrated into the same neuronal circuit?

A working model for the establishment and retrieval of punishment and safety memories

All the DANs of the mushroom body share a similar circuitry (**Fig. 9B**). They receive various input outside of the mushroom body neuropil, both from post-sensory interneurons and feedback pathways from the mushroom body output^{15,19,25}. The two major downstream partners of each DAN are the KCs of the mushroom body as well as the MBONs innervating the same compartment as the DAN^{6–7,16,19} (**Fig. 9 – Figure supplement 1A-C**). The behavioural effects of the DANs are most likely executed via the MBONs, as they are the only obvious connection from the DANs to motor control¹⁶. Most evidence, both from larval and adult *Drosophila*, suggest that memories are established via the DAN – KC synapses (see below). We suggest that, in turn, the DAN – MBON synapses may gate the memory retrieval (**Fig. 9B**). Notably, these two kinds of synapses are not spatially segregated. Rather, DAN – KC, DAN – MBON and KC – DAN synapses form several dense clusters on the DAN, with the DAN – KC and DAN – MBON synapses more to the centre, and the KC – DAN synapses more on the outside of each cluster (**Fig. 9 – Figure supplement 1D-E**). In many cases, one cluster contains synapses with several different MBONs, and several clusters contain synapses with the same MBON (**Fig. 9C**). We had demonstrated the same clustered pattern already for DAN-i1⁴⁶. The potential function of this arrangement remains speculative, but it may indicate that these clusters form functional units, each of them with all necessary synaptic connections for both memory formation and memory retrieval.

In the following, we use the d-compartment (lateral appendix) of the mushroom body as example. During training, the presence of an aversive US is signalled by DAN-d1 towards the mushroom body. The coincidence of dopaminergic input and activation of the KC by an odour during paired training is detected by the type I adenylate cyclase and results in presynaptic depression of the KC – MBON synapses (larvae^{6–7}; adults^{58–59}). This leads to less drive towards approach-promoting MBONs and hence to odour avoidance (**Fig. 9D**) (larvae¹⁶; adults^{18,60–62}). For the case of the larval d-compartment, MBON-d1 was shown to be indeed approach-promoting, MBON-d2 and d3 did not show any behavioural effect, and MBON-p1 has not been tested so far¹⁶. Thus, the KC – MBON-d1 synapses are the best candidates for harbouring a memory established by DAN-d1.

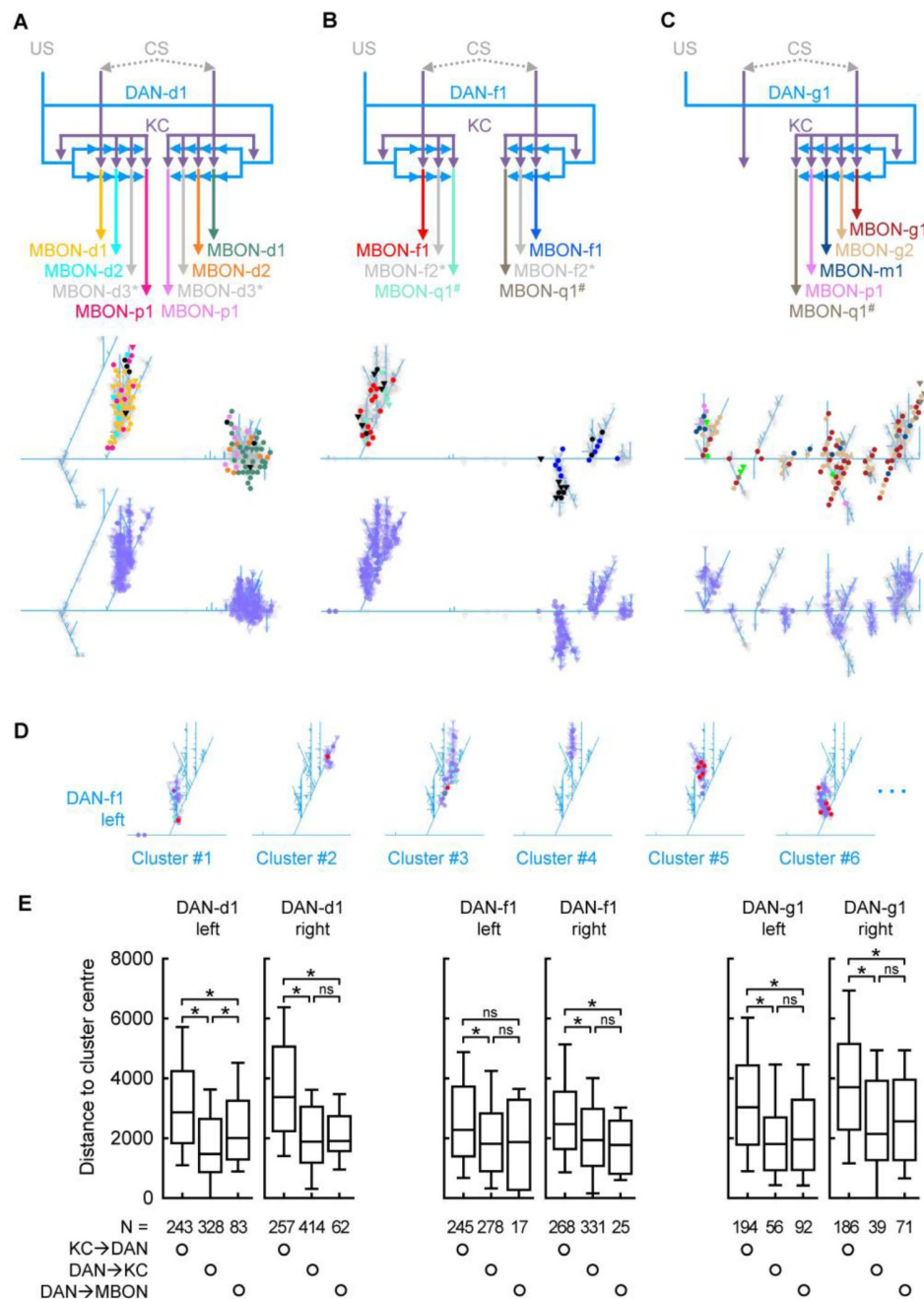


Figure 9 - supplement 1

Synaptic organization of the DL1-cluster DANs.

(A) Top: CS information is signalled towards the mushroom body Kenyon cells (KCs). DAN-d1 and most KCs on both hemispheres establish mutual synapses, and both DAN-d1 and KCs provide output to 4 MBONs. For simplicity, only synapses with the left DAN-d1 are shown. Middle: Truncated dendrogram of the left DAN-d1 based on EM and light microscopy^{19,53}. The positions of all synapses with MBONs are indicated in the colours shown in the top sketch. Bottom: The positions of all synapses with KCs are indicated in purple. (B-C) As in (A), but for DAN-f1 and DAN-g1, respectively. Note that DAN-g1 only innervates the contralateral mushroom body. *: this MBON has no synapses with the respective DAN in stage 1, but innervates the compartment in stage 3; #: this MBON establishes synapses with the respective DAN in stage 1 but is not found in stage 3^{19,53}. (D) The synapses are organized in clusters. Displayed are 6 sample clusters of the left DAN-f1. (E) Distances of all synapses within each cluster from the cluster centre. For all 6 DANs, KC → DAN synapses are further away from the centre than the other synapse types. * and ns above brackets indicate pairwise significance or non-significance, respectively (MW). For the underlying source data and the results of the statistical tests, see Supplementary Data 1. For high-resolution dendrograms of both the left and right DANs, see Supplementary Material S2.

Presenting an odour unpaired from DAN activation can lead to potentiation of the KC → MBON synapses¹⁷ and thus to odour approach (**Fig. 9E**). The details of this process are not well understood. We assume that during training signals from the DAN can have opposite effects on the KC → MBON synapse, depending on the activity state of the KC: if the DAN signal coincides with KC activation by an odour, the KC → MBON synapse is depressed (**Fig. 9D**), otherwise it is potentiated (**Fig. 9E**). This drives avoidance or approach towards the odour, respectively. In the case of DAN-d1, only the odour approach after unpaired training, indicating safety learning, was statistically significant (**Fig. 3B**).

During the test, the presence of the US is signalled by DAN-d1 and provides a gating signal. This signal is most likely received by the MBONs which are known to be necessary for the expression of learned behaviour (larvae^{16,53}; adults^{18,63–67}). Such a gating signal in the simplest case could be a logical ‘AND’ gate: for learned behaviour to occur, both the changed drive from the KC onto the MBON and the DAN gating signal to the MBON are required. The consequence is that any associative memory established by the modified KC → MBON-d1 synapses can only take effect behaviourally if DAN-d1 is activated also during the test, just as we observed (**Fig. 2B**).

How the mushroom body circuit can account for the modulations of acute locomotion

So far, very little is known about how the mushroom body-innervating DANs could modulate innate locomotion. We consider two simple scenarios to explain our observations, a reduced velocity and increased bending upon activation of DAN-d1 (**Fig. 8A–D**): (1) the DAN-d1 → MBON-d1 synapse acts inhibitory such that an activation of DAN-d1 silences the approach-promoting MBON-d1 (**Fig. 9F**); (2) the signal from the DAN-d1 → KC synapse reduces the drive from the KCs onto the MBON-d1 (**Fig. 9G**). In both cases, the net output of the mushroom body would be tilted towards aversion.

Scenario 1 seems straight-forward, but two studies in adults caution against such a simple model: First, it was found that activating a DAN while pharmacological blocking output from the KCs activated the MBON of the respective compartment rather than inhibiting it²⁰. So, we would need to assume that the “sign” of the DAN → MBON synapses differs either between developmental stages, or between different mushroom body compartments, or even changes in certain conditions. Different dopamine receptors can either excite or inhibit a neuron⁶⁸, and many DANs likely release additional neurotransmitters^{19,55,69}. Thus, it remains possible that activation of a DAN causes either activation or inhibition of the downstream MBON dependent on the exact composition of receptors.

Second, a recent study compared the innate valence induced by optogenetically activating DANs (of the PAM cluster) and MBONs, and found that in the case of DANs valence was brought about largely by differences in speed in darkness or light, but in the case of MBONs by modulations of turns at the dark-light border⁶⁹. This discrepancy makes a simple, direct inhibition unlikely.

Scenario 2 resembles the effects of the DAN signal in paired learning (**Fig. 9D,G**). However, the observed modulations of locomotion occurred within a few seconds and therefore much faster than in our learning experiments. In addition, some activation of the KCs and therefore some drive of the MBONs would be required, otherwise an acute down-regulation of the KC → MBON synapse would be without any effect. To which extent the KCs were active during our experiments is no known.

Previous studies provide conflicting evidence for the involvement of the KCs: on the one hand, a recent study in adults found that KC activity was dispensable for the innate attraction to PAM-DANs⁶⁹. On the other hand, the same study found that knocking down the Dop2R receptor in the

KCs reduced the innate attraction to PAM-DANs⁶⁹. In addition, a study in larvae found an increased locomotion when the Dop1R1 receptor was knocked-down in the KCs⁷⁰.

Considering these discrepancies, it is very likely that both scenarios are too simplistic, by ignoring for example asymmetric body-side effects of DAN and/or MBON activities⁴⁷, or cross-compartmental connections between DANs and MBONs¹⁵. It can also not be excluded that DANs exert their acute locomotion effects independent of the MBONs. Future research has to disentangle the neuronal mechanisms of how DAN activity modulates innate locomotion and valence.

Each DAN induces a different combination of effects

Strikingly, each DAN in this study (as well as in a previous study regarding the appetitive domain) induced a different set of behavioural effects (Fig. 9A).

The optogenetic activation of DAN-d1 and DAN-f1 can have two effects with respect to memories: during training, the virtual US signal they confer can induce a safety memory. During testing, this memory is only retrieved when the same DAN is activated (Fig. 2B-C). This mimics many natural US such as tastants or mild mechanical disturbance^{32–36}, or formic acid in adult flies⁷¹. We propose that this mechanism serves to ensure that memories only influence behaviour when it is adaptive to do so. In contrast, the activation of DAN-g1, as well as of two different combinations of DANs (which both happen to include DAN-g1), establish memories that can be retrieved also without the respective DANs being activated (Fig. 2D-F). This mimics US like electrical shocks that induce memories that are retrieved in absence of the US^{72–73}. It has been shown that memories in larvae can be specific for the kind of US and are retrieved independently³⁹. It is conceivable that memories regarding relatively harmless types of US, like a bad taste, require the US to be present in order to be retrieved and guide behaviour. Memories regarding more severe US which threaten the body integrity of the animal, such as electrical shocks, are retrieved without the actual US being present. The activation of DAN-d1 or DAN-f1 on the one hand, and DAN-g1 on the other hand, may correspond to such harmless and dangerous US, respectively.

In a previous study we found that the appetitive US-signalling DAN-i1 can establish memories of opposite valence when it was activated paired or unpaired with an odour⁴⁶. This suggested that both kinds of memory rely on the activity of the same DANs. In the aversive domain, DAN-d1, DAN-f1 and DAN-g1 each establish only safety memory but not punishment memory, whereas activating combinations of DANs did establish both kinds of memory¹⁵ (Fig. 3). The reason for these discrepancies is not known yet.

Most strikingly, DAN-d1 and DAN-g1 (but not DAN-f1) also modulate innate locomotion (Fig. 4, 8). Interestingly, recent studies in mammals suggested that the nigro-striatal ‘movement’ pathway also plays a role for carrying reward signals, in addition to the traditional mesolimbic ‘reward’ pathway⁷⁴. Together, these results point towards a neuronal organization in which the same dopaminergic neurons can have multiple functions that are as diverse as instructing learning processes and modulating movements.

Our results suggest a combinatorial-like neuronal organization with each neuron having a unique, yet overlapping set of functions. This raises a question: given that all mushroom body compartments share the same basic circuitry¹⁹ (Fig. 9B), how can such functional differences be explained? It is conceivable that they depend on the specific properties of the involved MBONs or its synapses with the KCs and the respective DANs. For example, if the synapses between the KCs and a given MBON are weak to begin with (for example the MBONs of the d- and f-compartment), they can be easily potentiated by unpaired training, but any depression by paired training will be minute. Such weak depression may need to happen in several compartments before it becomes behaviourally measurable.

Moreover, the impact of the gating signal by the DAN to the MBON may vary dependent on the circumstances: the memories established by DAN-d1 or DAN-f1 bypassed the retrieval control by the DANs if those memories were established with CsChrimson, but not with ChR2-XXL (**Fig. 2 – Figure supplement 3** [15](#)). The memory established by DAN-g1 bypassed this control even when using ChR2-XXL (**Fig. 4D** [4](#)). Little is known about the molecular mechanisms of memory retrieval in MBONs, and future research has to uncover which factors may determine its dependence on dopamine signals.

Our study revealed that individual DANs can induce surprisingly diverse effects, from establishing and retrieving memories to modulating innate locomotion. The mushroom body circuit – that is shared with adult flies and other insects – provides an excellent study case to advance our understanding of the complex functions of the dopaminergic system. Given the highly conserved roles of DANs in mediating “teaching” signals and modulating movement across the animal kingdom including vertebrates and humans, a deeper understanding of this circuit and its functions should provide new insights about the principles of DAN function in general.

Materials & Methods

Animals

Third-instar feeding-stage larvae (*Drosophila melanogaster*), aged 4-5 days after egg laying, were used throughout. Flies were maintained in mass culture at 25 °C, 60-70 % relative humidity and a 12/12 hour light/dark cycle. Two kinds of food medium were used:

Food medium #1 was used for all learning and olfactory experiments as well as for some of the locomotion experiments. Per 1 l water, 45 ml sugar beet molasses, 45 ml malt, 147.5 g cornmeal, 10 g soy flour, 18.75 g dry yeast, 6.25 g agar and 2.5 g methyl-4-hydroxybenzoate (Merck, Darmstadt, Germany) were added as described in detail previously [75](#).

Food medium #2 was used for most of the locomotion experiments as well as the olfactory stimulation preference experiments. Per 1 l water, 100g glucose, 50 g cornmeal, 31.65 g wheat germ, 40 g dry yeast, 8 g agar, 2 ml propionic acid (Kanto-kagaku, Tokyo, Japan) and 5 ml 10% butyl p-hydroxybenzoate (Wako, Osaka, Japan) in ethanol were added as described in detail previously [76](#).

We took a spoonful of food medium from a food vial, randomly selected the desired number of larvae, briefly rinsed them in tap water and started the experiment.

We used either wild-type larvae from the Canton Special genotype, or transgenic larvae to express light-gated ion channels in the respective DAN. Towards this end, one of two effector strains, *UAS-ChR2-XXL* (Bloomington Stock Center #58374), or *20xUAS-CsChrimson-mVenus* (Bloomington Stock Center #55134), was crossed to either *TH-Gal4* (Bloomington Stock Center #8848), *R58E02-Gal4* (Bloomington Stock #41347), or to one of four split-Gal4 driver strains: MB328B, MB054B, SS02180, or SS01716 [15](#) to obtain double-heterozygous offspring. As driver controls the driver strains were crossed to a local copy of *w¹¹¹⁸* (Bloomington Stock Center #3605, 5905, 6326). As effector control a strain carrying the landing sites used for the split-Gal4 (*attP40/attP2*), yet without a Gal4 domain inserted (“empty”) [77](#), was crossed to the respective effector. When using *TH-Gal4*, we used *w¹¹¹⁸* instead. As CsChrimson requires feeding of all-trans-retinal [54](#), larvae that were crossed to *20xUAS-CsChrimson-mVenus* were raised in standard food medium that additionally contained 400 µM final concentration of all-trans-retinal (CAS: 116-31-4, Sigma-Aldrich). For the TH-RNAi experiments, a double effector strain was used, made from the *UAS-ChR2XXL* strain and *y[1]v[1];P{TRiP.JF01813}attP2* (Bloomington Stock #25796) to express a double-strand-RNA element of the *ple* gene [78](#).

All larvae used for optogenetic experiments were raised in constant darkness. For the whole-body anatomy, we crossed each driver strain to a $y[1] w^{*}; MB247-mCherry-CAAX, UAS-mIFP-T2A-HO1$ double-effector strain²⁸ (called *MB247-mCherry-CAAX, UAS-mIFP* in the main text). As driver controls we crossed each driver strain to $y[1] w^{*}; MB247-mCherry-CAAX$ ²⁸, as effector control we crossed $y[1] w^{*}; MB247-mCherry-CAAX, UAS-mIFP-T2A-HO1$ to $y[1] w^{*}$ (Bloomington Stock Center #1495).

Experimental setup

For learning and preference experiments, larvae were trained and tested in Petri dishes of 9 cm inner diameter (order #82.1472, Sarstedt) filled with 1 % agarose (electrophoresis grade; CAS: 9012-36-6). For the experiments displayed in **Fig. 2A** and **3A**, 5 mM quinine (purity: 92%, CAS: 6119-70-6, Sigma-Aldrich) was added into the agarose. As odour, we used *n*-amyl acetate diluted 1:20 in paraffin oil (AM; CAS: 628-63-7, Merck). For locomotion experiments, either Petri dishes of 15 cm inner diameter (order #82.1184.500, Sarstedt) filled with 1 % agarose were used, or dishes of 9 cm inner diameter (order #3-1491-01, AS ONE) filled with 2 % agar (CAS: 9002-18-0, Fujifilm Wako Pure Chemical Industries).

Experiments were performed inside a 43 x 43 x 73 cm surrounding box equipped with a custom-made light table featuring a 24 x 12 LED array (Solarox) and a 6 mm thick diffusion plate of frosted plexiglass on top to ensure uniform blue light for ChR2-XXL activation (470 nm; 120 $\mu\text{W}/\text{cm}^2$; for **Fig. 2 – Figure supplement 3C**, **Fig. 4** incl. Figure supplements, **Fig. 7** incl. Figure supplements, **Fig. 8** and **Fig. 8 – Figure supplement 1**: 1000 $\mu\text{W}/\text{cm}^2$) or red light for CsChrimson activation (630 nm; 430 $\mu\text{W}/\text{cm}^2$). Petri dishes were placed directly on top of the diffusion plate. For additional details regarding the experimental setup, see⁵³.

Associative learning

Learning experiments followed established protocols⁷⁹. Odour containers were prepared by adding 10 μl of odour substance into custom-made Teflon containers (5 mm inner diameter with a lid perforated with 7 holes of 0.5-mm diameter each). Petri dishes were covered with modified lids perforated in the centre by 15 holes of 1 mm diameter each to improve aeration.

For paired training with quinine as US, about 20 larvae were placed on a quinine-filled Petri dish equipped with two AM-filled odour containers. After 2.5 min, the animals were transferred to a dish with plain agarose and empty odour containers (EM) on which they spent another 2.5 min. This cycle (AM-/EM) was repeated two more times before the animals were transferred to a test Petri dish with AM on one and EM on the other side. This dish was either a plain agarose dish, or one with quinine added, as mentioned in the results section. After three minutes, the number of animals on either side as well as on a 1 cm-wide middle stripe were counted and the odour preference was calculated as:

$$(1) \quad \text{Preference} = \frac{\#AM - \#EM}{\#Total}$$

Thus, preference values are constrained between 1 and -1 with positive values indicating a preference for and negative values indicating avoidance of AM. Note that the training sequence in every other repetition of the experiment was reversed (EM/AM-)

For unpaired training, AM was presented on plain agarose Petri dishes and EM was present on quinine Petri dishes, such that the CS and US were experienced explicitly separated from each other (AM/EM-or EM-/AM). Otherwise, the training was performed analogously, and the final test was performed as described.

From each one paired and one unpaired trained cohort of animals we calculated an associative Memory score as:

$$(2) \quad \text{Memory Score} = \frac{\text{Preference (Paired)} - \text{Preference (Unpaired)}}{2}$$

Thus, the values can range from 1 and -1 with negative values indicating that the larvae preferred the odour less after paired than unpaired training. Thus, negative scores indicate an aversive associative memory. Positive values, in turn, would indicate appetitive associative memory.

Learning experiments with ChR2-XXL or CsChrimson as effectors were performed analogously. However, all Petri dishes were filled with plain agarose, i.e. no tastants were added. Rather, for paired training, AM was paired with continuous blue (ChR2-XXL) or red light stimulation (CsChrimson) for 2.5 min, followed by 2.5 min of darkness without an odour. For unpaired training, animals received odour presentation and light stimulation separately from each other. The test was performed as described, either in darkness or with light stimulation as mentioned in the results section.

Innate olfactory behaviour

Odour containers and Petri dishes were prepared as described above. Cohorts of about 20 larvae were collected from the vial, briefly washed in tap water and placed to a Petri dish with an AM container on one side and an empty container on the other side. After 3 minutes, the innate odour preference was determined according to [equation \(1\)](#).

Locomotion experiments

Cohorts of about 8 larvae were placed on the middle of a Petri dish filled with agarose or agar. The dish was placed in the experimental chamber and video-recording was started after about 10 s. After 30 s of video-recording in darkness, light was switched on for 30 s followed by 60 s of darkness. In some experiments, 2 or 3 light stimulation cycles were applied. Blue or red light was used as mentioned along with the results.

Experiments involving feeding 3IY followed established protocols⁵⁷. Before the experiment the larvae were transferred to a yeast paste with or without added 3-Iodo-L-tyrosine (3IY; stored at -20°C; CAS: 70-78-0, Sigma-Aldrich) at a concentration of 5 mg/ml. After 4 hours, cohorts of 8 larvae were collected from the yeast paste and the experiment was conducted as described.

Optogenetic stimulation preference experiments

In order to test for the animals' preference for having sets of DANs activated, we prepared Petri dishes filled with agar of which one half was illuminated by blue light, the other half was kept in darkness. Cohorts of about 10 larvae were placed on the dark half of the dish about 1.5 cm from the midline. Their behaviour was video-recorded for 3 minutes before the animals were counted and the preference was calculated as:

$$(1) \quad \text{Light preference} = \frac{\# \text{Light} - \# \text{Darkness}}{\# \text{Total}}$$

Thus, preference values are constrained between 1 and -1 with positive values indicating a preference for and negative values indicating avoidance of the light.

Video analysis

Video analysis used the IMBA (Individual Maggot Behaviour Analyser) software and followed previous procedures⁵¹. In brief, the positions of head, tail, as well as several spine and contour points of each larva were tracked over the course of the experiment. A combination of computational modelling and statistical approaches allowed keeping the identities of larvae across

collisions and thus the analysis of individual animals tested in cohorts. Then, we discarded all tracks that were shorter than half of the time window used for the analysis. This prevented multiple tracks from the same individual animal from entering the analysis.

Larval behaviour was separated into head-casting (HC) phases and run phases. Lateral HCs are a very prominent feature during larval chemotaxis. We quantified these head movements based on the angular speed of the head vector through the front third of the animal. HCs were detected when the angular speed exceeded $\pm 35^\circ/\text{s}$. Runs were determined as periods in which the animals were not performing HCs, omitting 2 s before and after each HC event to prevent behavioural effects of the HC. We noticed that the rhythmic velocity changes of the tail point provided a reliable indicator for the occurrence of one peristaltic forward movement. We therefore defined each local maximum in the tail forward velocity as one step, and the period between two maxima, i.e. one complete peristaltic cycle, as an inter-step (IS) period. Since the local-maxima detection algorithm we employed also produced false maxima during intervals of little or no forward motion, we discarded maxima whenever the tail velocity in forward direction was less than 0.6 mm/s.

After the detection of runs, steps and head casts, we calculated a number of behavioural attributes. We summarize all attributes in this study in **table 1**. Whenever the average behaviour of a group of larvae is displayed over time, the mean and the 95 % confidence intervals of the respective attribute are calculated per 1 s time bin with a 5 s convolution filter. Whenever data are displayed in box plots to compare across experimental conditions, the mean of each attribute is calculated for each individual larva across the indicated time period. As we were interested in the difference in a given attribute X before and during light stimulation, we calculated Δ -values as the difference in X during the last 15 s of light stimulation minus X during the 15 s just before light stimulation (called baseline in the main text).

We used random forest models which are based on classification and regression trees^{80–81}. The random forest algorithm works by building N number of decision trees and then predicting the class label of an instance using a majority voting. Each tree is trained on a random subspace of the data, only using a sample of the instances. To evaluate the random forest classification, we used five-fold cross validation, which was repeated three times. Five-fold cross validation means that the test set for the model is taken from five different, disjunct sets of the data set. The data set is then shuffled and the procedure is repeated three times. Cross validation is used to make sure that the model is not based merely on the selected data, but can explain the full data set.

For the applications of the random forest in this study, we pre-selected 17 behavioural attributes to be included (**Table 1**). The pre-selection was based on our previous study⁵¹ and was warranted to avoid using largely redundant attributes and to exclude non-behavioural attributes such as the size of the larvae. The random forest was always applied to pairwise compare two experimental conditions, typically the experimental groups with one of the genetic controls. In order to provide a good estimate of the importance of each attribute, the complete random forest model (including the cross validation) was performed 10 times for any given comparison. In the typical case, this means 10 repetitions of the comparison between the experimental genotype with the driver control and 10 repetitions of the comparison with the effector control. In each case, the relative importance rank of each attribute was noted.

Whole-body anatomy

Experimental procedures follow previous work²⁸. In brief, third-instar larvae were first bleached (4% sodium hypochlorite, order #9062.1, Roth) for 10 min at room temperature. After washing (3 × 5 min in distilled water), they were fixed in 4% PFA in 0.1 mol/L PB at pH 9, with gentle shaking overnight at 4°C. Fixed samples were briefly rinsed 3 × with 0.1 mol/L PB containing 0.2% Triton X-100 (PBT), then washed 2 × 60 min and left overnight at 4°C in PBT. Dehydration the following day used a graded ethanol (EtOH) series (60 min in 10% and 25% EtOH,

followed by 30 min in 50%, 60%, 80%, and 2 × 100% EtOH). Samples were then cleared by replacing EtOH with ethyl cinnamate (ECi; ethyl 3-phenyl-2-propenoate; order #112372-100G, Sigma-Aldrich). After 30-60 min, the ECi was refreshed once, and the samples stored at room temperature in black boxes in a desiccator.

The samples were placed in ECi-cleared phytigel blocks (1 × 1 × 1 cm, P8169, Sigma-Aldrich) before imaging on an Ultra-Blaze microscope (Miltenyi Biotec). The microscope was equipped with an FIU-15 white light laser (NKT Photonics) for excitation through triple-sheet optics to illuminate samples from one side. The following excitation and emission filters were used: autofluorescence: exc. 480/20, em. 515/30, mcherry: exc. 590/20, em. 625/30, mIFP: exc. 680/20, em. 720/30.

Tiled image stacks were acquired with either a 12× objective with zoom set to 1 or 4x objective with zoom set to 1.66 (both Miltenyi Biotec) with a format of 2048 × 2048. Using ImSpector software (version 7.3.6, Miltenyi Biotec), a 10% overlap of tiles was set for stitching.

Image files were processed with Imaris software (version 10.1, Bitplane), including file conversion, stitching, further processing, and rendering. Either the ortho slicer, oblique slicer or clipping plane tool was used to restrict volumes in the z direction to improve the representation of structures that were otherwise covered in the context of the whole body. All inserts were generated using the snapshot function in Imaris. Whole-body 2D maximum-intensity projections were generated in Fiji⁸². Supplementary Movie 1 was produced in Imaris with the key frame animation tool; Adobe Premiere Pro 2020 (version 14.9.0, Adobe) was used for cutting and labelling.

Dendrogram analysis

To visualize neuron morphology and synapse locations, we employed neuron dendrograms⁸³ that are linearized, topologically correct 2D representations of 3D neuron reconstructions with synapse annotations (based on ¹⁹). Synapse locations and the distances between them are approximately correct, with slight adjustments to avoid overlap and cluttering of the synapse symbols⁸³.

We formally analyzed the spatial organization of synapses based on geodesic distances along the neuron's branches ("cable length distances"). The geodesic distance between two synapses was determined by finding the shortest path between them on the neuron, and then computing the cumulated Euclidean distance, in the 3D coordinate space of the neuron, between points along that path.

In particular, we analyzed spatially local clusters of synapses and the distances to the centroid of the cluster. To this end, we clustered the synapses on each neuron based on geodesic distances and using hierarchical clustering with Ward's criterion⁸⁴ (implemented in the base R package "stats"). The maximum number of clusters per neuron was set to k=10. For analysis, we kept only clusters with at least 5 presynaptic and 5 postsynaptic KC connections.

The centroid of a cluster was defined to be the most central synapse, computed by partitioning around medoids clustering (PAM with k = 1 cluster)⁸⁵ based on geodesic distances.

Quantification and statistical analysis

Two-tailed non-parametric tests were used. Values were compared across multiple groups with Kruskal-Wallis tests (KW). In case of significance, subsequent pair-wise comparisons used Mann-Whitney U-tests (MW). To test whether values of a given group differ from zero, we used Wilcoxon Signed-Rank tests (WSR). When multiple comparisons were performed within one analysis, a Bonferroni-Holm correction was applied to keep the experiment-wide error rate below 5 %⁸⁶. The detailed results of all statistical tests are reported in the Supplementary Data S1 along with the source data. We present our data as box plots which represent the median as the middle line and

25 %/75 % and 10 %/90 % as box boundaries and whiskers, respectively. Outliers are not displayed. Experimenters were blind to genotype. Sample sizes (biological replications) were chosen based on previous studies revealing moderate to mild effect sizes⁴⁷[47](#),⁵¹[51](#) and are displayed in the figures.

Data availability

Source data to all behavioural experiments are included as supplementary material (Supplementary Data S1).

Acknowledgements

This study received institutional support by the Otto von Guericke Universität Magdeburg, the Wissenschaftsgemeinschaft Gottfried Wilhelm Leibniz (WGL), the Leibniz Institute for Neurobiology (LIN), the Hokkaido University, and the RWTH University Aachen. This study received grant support from the Deutsche Forschungsgemeinschaft (DFG) (GE 1091/4-1, and FOR 2705 *Mushroom body*), the Japan Society for the Promotion of Science (JSPS) (Kakenhi 22K20673) and the German-Israeli Foundation for Science (GIF) (G-2502-418.13/2018). Experimental contributions of J. Chen, R. Khondaker, T. Meyer, S. Schuller, M. Thane, L. Warzog, B. Fischer, technical assistance by M. Dombach, M. Paisios, M. Thane and F. Unterstab, as well as discussions with C. Eschbach, B. Gerber, C. König, N. Mancini, N. Tanaka, A. Thum and A. Yarali are gratefully acknowledged. We thank C. Eschbach, C. König, A. Thum, D. Weber and M. Zlatić for sharing fly strains and results prior to publication.

Additional information

Author contributions

Conceptualization: Michael Schleyer, Naoko Toshima.

Formal analysis: Michael Schleyer, Arman Behrad, Franziska Behnke.

Investigation: Naoko Toshima, Arman Behrad, Franziska Behnke, Gauri Kaushik, Aliće Weiglein, Martin

Strauch, Juliane Thoener, Oliver Kobler, Maia Lisandra M. Wang, Markus Dörr.

Data curation: Michael Schleyer, Naoko Toshima, Arman Behrad, Aliće Weiglein, Franziska Behnke.

Writing - original draft: Michael Schleyer, Naoko Toshima, Aliće Weiglein, Franziska Behnke.

Writing - review & editing: Naoko Toshima, Aliće Weiglein, Arman Behrad, Franziska Behnke, Gauri Kaushik, Martin Strauch, Juliane Thoener, Oliver Kobler, Maia Lisandra M. Wang, Markus Dörr.

Visualization: Michael Schleyer, Martin Strauch, Juliane Thoener, Oliver Kobler.

Supervision: Michael Schleyer, Juliane Thoener.

Project administration: Michael Schleyer.

Funding acquisition: Michael Schleyer.

Competing financial interests

The authors declare no competing financial interests.

Additional files

Supplementary Data S1. Raw data and results of statistical tests. This excel file provides all raw data underlying the figures. Each tab shows the data of one figure or supplementary figure, with the results of all statistical tests to the left. [🔗](#)

Supplementary Material S1. Full body expression of all used driver strains and genetic controls. Each page shows the three channels (autofluorescence, mCherry, mIFP) for the experimental genotype (driver strain crossed to MB247-mCherry-CAAX, UAS-mIFP) and a driver control lacking the UAS construct for one driver strain. On the first page, in addition the effector control crossed to *yw* is displayed. Next to the whole body, the insert shows a volume-restricted magnification of the CNS region. Blue arrowheads indicate DAN cell bodies. Scale bars represent 250 μm for the full body and 50 μm for the inserts. [🔗](#)

Supplementary Material S2. High-resolution dendrograms. Each page shows the dendrogram of one DAN in high resolution. For clarity, the synapses with KCs and other synapses are displayed separate. Even pages highlight only synapses with MBONs, odd pages synapses with KCs. All other synapses are represented with transparent grey symbols. Circles display synapses from the given DAN to other neurons, triangles synapses from other neurons to that DAN. The colour code is given on each page. [🔗](#)

Supplementary Movie 1. An animation through the data set used in Fig. 1E. TH-Gal4 was crossed to MB247-mCherry-CAAX, UAS-mIFP, and both the mCherry-CAAX (magenta) and the mIFP (yellow) signals are shown. The movie starts with the larva's head towards the front left. The second half shows the animal from top. At the end, the z-layers covered by the proventriculus and the melanin-producing cells were removed to allow the cells of the CNS to become visible. Grid spacing: 200 μm . [🔗](#)

Supplementary Movie 2. A sample video of the experiment presented in Fig. 4 with TH/ChR2-XXL larvae. The video is played in double speed. Blue light (not visible in the video) was switched on at timestamp 15 s (30 s in the real experiment) and off at timestamp 30 s (60 s in the real experiment). Larval heads are marked in green, tails in red. [🔗](#)

Supplementary Movie 3. A sample video of the experiment presented in Fig. 4 with R58E02/ChR2-XXL larvae. The video is played in double speed. Blue light (not visible in the video) was switched on at timestamp 15 s (30 s in the real experiment) and off at timestamp 30 s (60 s in the real experiment). [🔗](#)

Supplementary Movie 4. A sample video of the experiment presented in Fig. 6 with TH/ChR2-XXL larvae. The video is played in double speed. The right half of the dish was illuminated with blue light (not visible in the video) throughout the experiment. [🔗](#)

Supplementary Movie 5. A sample video of the experiment presented in Fig. 6 with R58E02/ChR2-XXL larvae. The video is played in double speed. The right half of the dish was illuminated with blue light (not visible in the video) throughout the experiment. [🔗](#)

References

1. Schultz W. (2015) **Neuronal reward and decision signals: from theories to data** *Physiol Rev* **95**:853–951
2. Kaun K.R., Rothenfluh A. (2017) **Dopaminergic rules of engagement for memory in *Drosophila*** *Curr Op Neurobiol* **43**:56–62
3. Gershman S. J., Uchida N. (2019) **Believing in dopamine** *Nat Rev Neurosci* **20**:703–714
4. Speranza L., di Porzio U., Viggiano D., de Donato A., Volpicelli F. (2021) **Dopamine: the neuromodulator of long-term synaptic plasticity, reward and movement control** *Cells* **10**:735
5. Ryczko D., Dubuc R. (2023) **Dopamine control of downstream motor centers** *Curr Opin Neurobiol* **83**:102785
6. Widmann A., Eichler K., Selcho M., Thum A.S., Pauls D. (2018) **Odour-taste learning in *Drosophila* larvae** *J Insect Physiol* **106**:47–54
7. Thum A.S., Gerber B. (2019) **Connectomics and function of a memory network: the mushroom body of larval *Drosophila*** *Curr Op Neurobiol* **54**:146–154
8. Boto T., Tomchik S. M. (2020) **Cellular and circuit mechanisms of olfactory associative learning in *Drosophila*** *J Neurogenet* **34**:36–46
9. Lammel S., Lim B. K., Malenka R. C. (2014) **Reward and aversion in a heterogeneous midbrain dopamine system** *Neuropharmacology* **76**:351–9
10. Groessl F., et al. (2018) **Dorsal tegmental dopamine neurons gate associative learning of fear** *Nat Neurosci* **21**:952–962
11. Menegas W., Akiti K., Amo R., Uchida N., Watabe-Uchida M. (2018) **Dopamine neurons projecting to the posterior striatum reinforce avoidance of threatening stimuli** *Nat Neurosci* **21**:1421–1430
12. de Jong J. W., et al. (2018) **A neural circuit mechanism for encoding aversive stimuli in the mesolimbic dopamine system** *Neuron* **101**:133–151
13. Kutlu M. G., Tat J., Christensen B. A., Zachry J. E., Calipari E.S. (2023) **Dopamine release at the time of a predicted aversive outcome causally controls the trajectory and expression of conditioned behavior** *Cell Rep* **42**:112948
14. Modi M. N., Shuai Y., Turner G. C. (2020) **The *Drosophila* mushroom body: from architecture to algorithm in a learning circuit** *Annu Rev Neurosci* **43**:465–484
15. Eschbach C. (2020) **Recurrent architecture for adaptive regulation of learning in the insect brain** *Nat Neurosci* **23**:544–555

16. Eschbach C. (2021) **Circuits for integrating learned and innate valences in the insect brain** *eLife* **10**:e62567 <https://doi.org/10.7554/eLife.62567>
17. Cohn R., Morante I., Ruta V. (2015) **Coordinated and compartmentalized neuromodulation shapes sensory processing in *Drosophila*** *Cell* **163**:1742–1755
18. Oswald D., et al. (2015) **Activity of defined mushroom body output neurons underlies learned olfactory behaviour in *Drosophila*** *Neuron* **86**:417–427
19. Eichler K., et al. (2017) **The complete connectome of a learning and memory centre in an insect brain** *Nature* **548**:175–182
20. Takemura S.Y., et al. (2017) **A connectome of a learning and memory center in the adult *Drosophila* brain** *eLife* **6**:e26975 <https://doi.org/10.7554/eLife.26975>
21. Menzel R. (2001) **Searching for the memory trace in a mini-brain, the honeybee** *Learn Mem* **8**:53–62
22. Strausfeld N.J., Sinakevitch I., Brown S.M., Farris S.M. (2009) **Ground plan of the insect mushroom body: functional and evolutionary implications** *J Comp Neurol* **513**:265–291
23. Schürmann F.W. (2016) **Fine structure of synaptic sites and circuits in mushroom bodies of insect brains** *Arthropod Struct Dev* **45**:399–421
24. Weiglein A., et al. (2021) **Aversive teaching signals from individual dopamine neurons in larval *Drosophila* show qualitative differences in their temporal “fingerprint”** *J Comp Neurol* **529**:1553–1570
25. Weber D., Vogt K., Miroshnikov A., Pankratz M, Thum A. S. (2023) **Four individually identified paired dopamine neurons signal taste punishment in larval *Drosophila*** *eLife* **12**:RP91387 <https://doi.org/10.7554/eLife.91387.1>
26. Selcho M., Pauls D., Han K.A., Stocker R.F., Thum A.S. (2009) **The role of dopamine in *Drosophila* larval classical olfactory conditioning** *PloS one* **4**:e5897
27. Rohwedder A., et al. (2016) **Four individually identified paired dopamine neurons signal reward in larval *Drosophila*** *Curr Biol* **26**:661–669
28. Kobler O., Weiglein A., Hartung K., Chen Y.-C., Gerber B., Thomas U. (2021) **A quick and versatile protocol for the 3D visualization of transgene expression across the whole body of larval *Drosophila*** *J Neurogenet* **35**:306–319
29. Neckameyer W. S., White K. (1993) ***Drosophila* tyrosine hydroxylase is encoded by the pale locus** *J Neurogenet* **8**:189–199
30. Verlinden H. (2018) **Dopamine signalling in locusts and other insects** *Insect Biochem Mol Biol* **97**:40–52
31. Erhardt B., et al. (2023) **Downregulation of Plasma Membrane Ca²⁺ ATPase driven by tyrosine hydroxylase-Gal4 reduces *Drosophila* lifespan independently of effects in neurons** *Fly (Austin)* **17**:2192457
32. Gerber B., Hendel T. (2006) **Outcome expectations drive learned behaviour in larval *Drosophila*** *Proc Biol Sci* **273**:2965–2968

33. Niewalda T., Singhal N., Fiala A., Saumweber T., Wegener S., Gerber B. (2008) **Salt processing in larval *Drosophila*: choice, feeding, and learning shift from appetitive to aversive in a concentration-dependent way** *Chem Senses* **33**:685–92
34. Eschbach C., et al. (2011) **Associative learning between odorants and mechanosensory punishment in larval *Drosophila*** *J Exp Biol* **214**:3897–905
35. El-Keredy A., Schleyer M., König C., Ekim A., Gerber B. (2012) **Behavioural analyses of quinine processing in choice, feeding and learning of larval *Drosophila*** *PLoS One* **7**:e40525
36. Toshima N., Kantar Weigelt M., Weiglein A., Boetzel F. A., Gerber B. (2019) **An amino-acid mixture can be both rewarding and punishing to larval *Drosophila melanogaster*** *J Exp Biol* **222**:jeb209486
37. Schleyer M., et al. (2011) **A behaviour-based circuit model of how outcome expectations organize learned behaviour in larval *Drosophila*** *Learn Mem* **18**:639–653
38. Schleyer M., Diegelmann S., Michels B., Saumweber T., Gerber B. (2013) **'Decision-making' in larval *Drosophila*** In: Menzel R., Benjamin P., editors. *Invertebrate Learning and Memory* München: Elsevier pp. 41–55
39. Schleyer M., Miura D., Tanimura T., Gerber B. (2015) **Learning the specific quality of taste reinforcement in larval *Drosophila*** *eLife* **4**:e04711 <https://doi.org/10.7554/eLife.04711>
40. Paisios E., Rjosk A., Pamir E., Schleyer M. (2017) **Common microbehavioural “footprint” of two distinct classes of conditioned aversion** *Learn Mem* **24**:191–198
41. Dawydow A., et al. (2014) **Channelrhodopsin-2-XXL, a powerful optogenetic tool for low-light applications** *Proc Natl Acad Sci USA* **111**:13972–13977
42. von Essen A. M. H. J., Pauls D., Thum A. S., Sprecher S. G. (2011) **Capacity of visual classical conditioning in *Drosophila* larvae** *Behav Neurosci* **125**:921–929
43. Toshima N., Schleyer M. (2024) **IR76b-expressing neurons in *Drosophila melanogaster* are necessary for associative reward learning of an amino acid mixture** *Biol Lett* **20**:20230519
44. Rescorla R. A. (1967) **Pavlovian conditioning and its proper control procedures** *Psychol Rev* **74**:71–80
45. Malaka R. (1999) **Models of classical conditioning** *Bull Math Biol* **61**:33–83
46. Schleyer M., Fendt M., Schuller S., Gerber B. (2018) **Associative learning of stimuli paired and unpaired with reinforcement: evaluating evidence from maggots, flies, bees, and rats** *Front Psychol* **9**:1494
47. Schleyer M., et al. (2020) **Identification of dopaminergic neurons that can both establish associative memory and acutely terminate its behavioral expression** *J Neurosci* **40**:5990–6006
48. Sen E., et al. (2024) **Cognitive limits of larval *Drosophila*: testing for conditioned inhibition, sensory preconditioning, and second-order conditioning** *Learn Mem* **31**:a053726
49. Jacob P. F., Waddell S. (2020) **Spaced training forms complementary long-term memories of opposite valence in *Drosophila*** *Neuron* **106**:977–991

50. Zhao B., Sun J., Li Q., Zhong Y. (2021) **Differential conditioning produces merged long-term memory in *Drosophila*** *eLife* **10**:e66499 <https://doi.org/10.7554/eLife.66499>
51. Thane M., et al. (2023) **High-resolution analysis of individual *Drosophila melanogaster* larvae uncovers individual variability in locomotion and its neurogenetic modulation** *Open Biol* **13**:220308
52. Xiang Y., Yuan Q., Vogt N., Looger L. L., Yeh Jan L., Nung Jan Y. (2010) **Light-avoidance-mediating photoreceptors tile the *Drosophila* larval body wall** *Nature* **468**:921–926
53. Saumweber T., et al. (2018) **Functional architecture of reward learning in mushroom body extrinsic neurons of larval *Drosophila*** *Nat Comm* **9**:1104
54. Klapoetke N. C., et al. (2014) **Independent optical excitation of distinct neural populations** *Nat Methods* **11**:338–46
55. Aso Y., et al. (2019) **Nitric oxide acts as a cotransmitter in a subset of dopaminergic neurons to diversify memory dynamics** *eLife* **8**:e49257 <https://doi.org/10.7554/eLife.49257>
56. Neckameyer W. S. (1996) **Multiple roles for dopamine in *Drosophila* development** *Dev Biol* **176**:209–19
57. Thoener J., et al. (2021) **Associative learning in larval and adult *Drosophila* is impaired by the dopamine-synthesis inhibitor 3-Iodo-L-tyrosine** *Biol Open* **10**:bio058198
58. Cognigni P., Felsenberg J., Waddell S. (2018) **Do the right thing: neural network mechanisms of memory formation, expression and update in *Drosophila*** *Curr Op Neurobiol* **49**:51–58
59. Tumkaya T., Ott S., Claridge-Chang A. (2018) **A systematic review of *Drosophila* short-term-memory genetics: Meta-analysis reveals robust reproducibility** *Neurosci Biobehav Rev* **95**:361–382
60. Aso Y., et al. (2014) **Mushroom body output neurons encode valence and guide memory-based action selection in *Drosophila*** *eLife* **3**:e04580 <https://doi.org/10.7554/eLife.04580>
61. Hige T., Aso Y., Modi M.N., Rubin G.M., Turner G.C. (2015) **Heterosynaptic plasticity underlies aversive olfactory learning in *Drosophila*** *Neuron* **88**:985–998
62. Felsenberg J., et al. (2018) **Integration of parallel opposing memories underlies memory extinction** *Cell* **175**:709–722
63. Sejourne J., et al. (2011) **Mushroom body efferent neurons responsible for aversive olfactory memory retrieval in *Drosophila*** *Nat Neurosci* **14**:903–910
64. Placais P.Y., Trannoy S., Friedrich A.B., Tanimoto H., Preat T. (2013) **Two pairs of mushroom body efferent neurons are required for appetitive long-term memory retrieval in *Drosophila*** *Cell Rep* **5**:769–780
65. Bouzaiane E., Trannoy S., Scheunemann L., Placais P.Y., Preat T. (2015) **Two independent mushroom body output circuits retrieve the six discrete components of *Drosophila* aversive memory** *Cell Rep* **11**:1280–1292
66. Shyu W.H., et al. (2017) **Neural circuits for long-term water-reward memory processing in thirsty *Drosophila*** *Nat Comm* **8**:15230

67. Wu J.K., Tai C.Y., Feng K.L., Chen S.L., Chen C.C., Chiang A.S. (2017) **Long-term memory requires sequential protein synthesis in three subsets of mushroom body output neurons in *Drosophila*** *Sci Rep* **7**:7112
68. Caline S, Karam C. K., Jones S. K., Javitch J. A. (2019) **Come fly with me: An overview of dopamine receptors in *Drosophila melanogaster*** *Basic Clin Pharmacol Toxicol* **126**:56–65
69. Mohammad F., Mai Y., Ho J., Zhang X., Ott S., Stewart J. C., Claridge-Chang A. (2024) **Dopamine neurons that inform *Drosophila* olfactory memory have distinct, acute functions driving attraction and aversion** *PLoS Biol* **22**:e30028432024
70. Silva B., Hidalgo S., Campusano J. M. (2020) **Dop1R1, a type 1 dopaminergic receptor expressed in mushroom bodies, modulates *Drosophila* larval locomotion** *PLoS One* **15**:e0229671
71. Schnaitmann C., Vogt K., Triphan T., Tanimoto H. (2010) **Appetitive and aversive visual learning in freely moving *Drosophila*** *Front Behav Neurosci* **4**:10
72. Khurana S., Bin Abu Baker M., Siddiqi O. (2009) **Odour avoidance learning in the larva of *Drosophila melanogaster*** *J Biosci* **34**:621–631
73. Pauls D., Pfizenmaier P. E. R., Krebs-Wheaton R., Selcho M., Stocker R. F., Thum A. S. (2010) **Electric shock-induced associative olfactory learning in *Drosophila* larvae** *Chem Senses* **35**:335–46
74. Wise R. A. (2009) **Roles for nigrostriatal--not just mesocorticolimbic--dopamine in reward and addiction** *Trends Neurosci* **32**:517–524
75. Michels B., et al. (2018) **Memory enhancement by ferulic acid ester across species** *Sci Adv* **4**:eaat6994
76. Sugime Y., Watanabe D., Yasuno Y., Shinada T., Miura T., Tanaka N. K. (2017) **Upregulation of juvenile hormone titers in female *Drosophila melanogaster* through mating experiences and host food occupied by eggs and larvae** *Zoolog Sci* **34**:52–57
77. Pfeiffer B.D., et al. (2010) **Refinement of tools for targeted gene expression in *Drosophila*** *Genetics* **186**:735–755
78. König C., Khalili A., Ganesan M., Nishu A. P., Garza A. P., Niewalda T., Gerber B., Aso Y., Yarali A. (2018) **Reinforcement signaling of punishment versus relief in fruit flies** *Learn Mem* **25**:247–257
79. Michels B., et al. (2017) **Pavlovian conditioning of larval *Drosophila*: an illustrated, multilingual, hands-on manual for odour-taste associative learning in maggots** *Front Behav Neurosci* **11**:45
80. Mola F. (1998) **Classification and Regression Trees Software and New Developments** In: Rizzi A., Vichi M., Bock H.-H., editors. *Advances in Data Science and Classification* Berlin, Heidelberg: Springer Berlin Heidelberg pp. 311–318
81. Breiman L. (2001) **Random Forests** *Machine Learning* **45**
82. Schindelin J., et al. (2012) **Fiji: an open-source platform for biological-image analysis** *Nat Methods* **9**:676–682

83. Strauch M., Hartenstein V., Andrade I.V., Cardona A., Merhof D. (2018) **Annotated dendrograms for neurons from the larval fruit fly brain** In: Puig Puig A., Schultz T., Vilanova A., editors. *Eurographics Workshop on Visual Computing for Biology and Medicine*
84. Aggarwal C.C., Reddy C. K. (2014) **Data Clustering: Algorithms and Applications** Chapman and Hall/CRC
85. Kaufman L., Rousseeuw P. J. (2005) **Finding groups in data: an introduction to cluster analysis** John Wiley & Sons
86. Holm S. (1979) **A simple sequentially rejective multiple test procedure** *Scand J Statist* **6**:65–70

Author information

Naoko Toshima

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany, Hokkaido University, Institute for the Advancement of Higher Education, Sapporo, Japan
ORCID iD: [0000-0001-9245-7494](https://orcid.org/0000-0001-9245-7494)

Arman Behrad[†]

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany

[†]Current address: Technical University Dresden, Computational Neuroscience, Department of Child and Adolescent Psychiatry, Faculty of Medicine, Dresden, Germany

Franziska Behnke

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany

Gauri Kaushik

Hokkaido University, Institute for the Advancement of Higher Education, Sapporo, Japan, Hokkaido University, Graduate School of Life Science, Sapporo, Japan

Alić Weiglein[‡]

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany
ORCID iD: [0000-0003-4626-328X](https://orcid.org/0000-0003-4626-328X)

[‡]Current address: Otto-von Guericke University Magdeburg, Institute for Biometry and Medical Informatics, Magdeburg, Germany

Martin Strauch[¶]

RWTH Aachen University, Imaging and Computer Vision, Aachen, Germany

[¶]Current address: Institute of Pathology, University Clinic Aachen, RWTH Aachen University, Germany

Juliane Thoener

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany

Oliver Kobler

Leibniz Institute for Neurobiology (LIN), Combinatorial NeuroImaging Core Facility, Magdeburg, Germany
ORCID iD: [0000-0001-7173-9919](https://orcid.org/0000-0001-7173-9919)

Maia Lisandra M Wang

Hokkaido University, Institute for the Advancement of Higher Education, Sapporo, Japan, Hokkaido University, School of Science, Sapporo, Japan

Markus Dörr*

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany

*Current address: Technical University of Applied Sciences Würzburg-Schweinfurt, Würzburg, Germany

Michael Schleyer

Leibniz Institute for Neurobiology (LIN), Department Genetics of Learning and Memory, Magdeburg, Germany, Hokkaido University, Institute for the Advancement of Higher Education, Sapporo, Japan
ORCID iD: [0000-0001-8020-1483](https://orcid.org/0000-0001-8020-1483)

For correspondence: m.schleyer@oia.hokudai.ac.jp

Editors

Reviewing Editor

Meet Zandawala

University of Nevada Reno, Reno, United States of America

Senior Editor

Albert Cardona

University of Cambridge, Cambridge, United Kingdom

Reviewer #1 (Public review):

Summary:

The authors investigate the role of different specific dopaminergic neurons in the mushroom body of *Drosophila* larvae for learning and innate behavior. All the tested neurons are thought to be involved in punishment learning. The authors discover that artificial activation of single DANs in training leads to safety learning, but not punishment learning. Furthermore, activation of single DANs can lead to changes in locomotion behavior, which can affect light preference. The authors provide a deeper understanding of the functional

diversity of single dopamine neurons; however, it is unclear how translatable these findings are to learning experiments with real punishment stimuli.

Strengths:

The authors attempt to disentangle what kind of memories are formed with the activation of different dopamine neurons - safety learning, and punishment learning, will the US be required to test for recall or not? They do indeed find differences and the results will be of interest to the learning and memory community.

Interestingly, optogenetic activation of a single DAN during training leads to safety memory, but not punishment memory. Furthermore, DAN activation also affects innate locomotion, and the authors can show that optogenetic activation of different DANs affects locomotion differently.

Weaknesses:

All experiments in the manuscript use optogenetic activation of DANs, thus it is not clear what kind of memories are formed. Several stimuli can be used as punishment, such as electric shock, salt, bitter, and light - it is not clear what kind of memory the authors investigate here. The findings could be discussed in the context of what DANs respond to. Furthermore, studies in adults and larvae showed that most DANs can code for both valences - etc., aversive DANs can be activated by punishment, and inhibited by reward. Thus, safety learning might be a result of a decrease in activity in DANs during odor presentation. The authors also do not discuss possible feedback loops from MBONs to DANs across compartments. Could such connections allow for safety learning in larvae?

The authors show that artificial activation with different light intensities can form different memories and that increasing the light intensity sometimes leads to no memories. Also, using different optogenetic tools reveals different results. This again raises the question of how applicable the results will be for learning with real stimuli. Is there a natural stimulus that only induces safety learning, but no punishment learning?

The authors provide a detailed behavioral analysis of locomotion behavior; however, the detailed analysis seems unnecessary for that dataset. Modulation of speed and bending rate has been described before with simpler methods (specifically for MBONs). The revealed locomotion phenotypes probably affect larval locomotion during memory recall with light activation, thus the authors should show that larvae are potentially able to move during light-on memory tests.

<https://doi.org/10.7554/eLife.106148.1.sa2>

Reviewer #2 (Public review):

Summary:

This study provides valuable context for ongoing research on the role of dopamine in memory and locomotion. DANs have been a fascinating area of study due to their complexity, and this work dissects specific DANs, exploring their roles in different memory-related behaviors while offering some explanations. The discussions provided by the authors effectively situates the study in the broader field of learning, memory, DAN circuitry and behavioral computation in insect brains. The study achieves what it sets out to and it does so unequivocally. The experiments were elegantly designed, leaving little room for doubt in the study's claims. However, the study lacks context regarding the molecular pathways underlying these results. While it strengthens current knowledge by providing robust evidence, it does little to explore the molecular mechanisms behind these effects.

Strengths:

(1) Experiment design is one of the strengths of this study. The experiments are thorough and cover the length and breadth of the core findings of the study. Although a lot of work has already been done in studying the role of dopamine in memory and locomotion, the dissection of the functions of distinct DANs in larvae has been done meticulously with well-structured experiments.

(2) This study fits quite nicely into the puzzle of memory, especially in the context of Dopamine. Previous studies in *Drosophila* adults have shown the opposing roles of DANs in locomotion depending on the context of DAN activation. This study drives that point home for larvae, providing conclusive evidence in that regard.

(3) The use of clear figures and simple language is one of the strengths of this paper. The figures are comprehensive, complete and manage to narrate the story by themselves. The flow of information is smooth. The simple and effective language used maintains scientific rigor while remaining accessible to those new to the field. A pleasant read.

Weaknesses:

(1) The authors have done a great job at structuring the figures. But some main figures would benefit from including the controls instead of placing them in supplementary.

(2) The paper would benefit from a deeper discussion regarding molecular mechanisms underlying their results. It would be interesting to see what the authors think about different Dopamine receptors and how they relate to the findings of this paper.

(3) Throughout the paper, the authors have been clear and comprehensive, but in some cases, further explanation of their choices were missing. For example, the choice to compare bending and tail velocity over other parameters within the same clusters is unclear.

<https://doi.org/10.7554/eLife.106148.1.sa1>

Reviewer #3 (Public review):

Summary

Across species, dopamine release carries out seemingly diverse functions, like reinforcing memories and regulating locomotion and flight. However, whether distinct dopaminergic neurons (DANs) are allocated for each function is not clear. In this study, Toshima et al. have used the numerically simple organization of the *Drosophila* larval brain to answer this question. They use optogenetic activation to systematically stimulate a small set of DANs, individually and collectively, and study the effect on diverse functions such as memory formation, retrieval, and locomotion. They find that singly or collectively, DL1 DANs can induce punishment and/or safety memory formation and retrieval. DANs can even gate the expression of memory. Finally, the same DANs also modulate locomotion in the larvae. The authors speculate that dopaminergic neurons in other species may also share such overlapping functions. Their findings are nicely summarised in Figure 9.

Strengths

The study comprehensively activates the neurons in the DL1 cluster in a systematic manner. Individual and collective stimulation of the DL1 DANs has been conducted to assess the induction and gating of aversive punishment memory, safety memory, and acute locomotion.

Specific adult *Drosophila* DANs are known to induce dual behaviors and functions. The same MP1/y1pedc DANs are recognized for gating appetitive memory expression and representing aversive teaching signals downstream of sensory stimuli such as electric shocks, bitter tastes, and heat. Neurons in the PPL1 cluster regulate adult flight and food-seeking behavior. The authors deserve credit for conducting an organized examination of dopaminergic neuron

functions in larvae, which makes their findings more comparable and facilitates the proposal of a holistic model.

They have provided substantial evidence for their findings and frequently presented replicated behavioral data sets. They have been transparent about results that were difficult to explain. Additionally, they have provided an impressive body of supporting data to strengthen their main findings.

Weaknesses

The larvae exhibit directed locomotory action to express punishment or safety memory. If the larvae did not move, we would not be able to assess memory function. Hence, functional activation of DANs could result in one action, which seems like two different functions of memory expression and locomotion. It can also be argued that activation of DANs represents a teaching signal to the KCs, and then eventually, downstream of the MBONs, it results in locomotion modulation. Hence, the seeming functional diversity could be a function of different downstream neuronal pathways and not molecular context-dependent diversity inside dopaminergic neurons. The authors should address this possibility or point out the fallacy in the above argument.

The finding that activation of TH-GAL4 conveys aversive valence and R58E02-GAL4 conveys appetitive valence seems redundant (Figure 6). I understand they say this in the context of locomotion. However, they may not have mentioned similar findings in adults. In adults, artificial activation of DANs covered by the same GAL4 lines acts as aversive and appetitive teaching signals for memory formation. These references should be cited appropriately in the results and discussion if not currently included.

The evidence for the role of dopamine (Figure 7) can be bolstered by using other available RNAi lines against TH. A valium20 vector-based shRNA line is recommended. The current evidence is based mainly on non-specific pharmacological intervention with 3IY.

<https://doi.org/10.7554/eLife.106148.1.sa0>